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CSGC: Cross-Stage Geometric Conditioning for Oriented Object Detection in Remote Sensing Images

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***Abstract*—Oriented object detection in remote-sensing imagery is complicated by dense spatial layouts, arbitrary orientations, and pronounced aspect-ratio variation. Although geometric cues can inform feature representation, region-of-interest (RoI) sampling, and training assignment, their joint treatment across these interfaces remains underexplored. We introduce cross-stage geometric conditioning (CSGC), a coordinated design spanning all three interfaces in a two-stage oriented detector. Cross-scale rotation-conditioned channel–spatial coupling (CSSC) aggregates rotated responses across pyramid levels to condition channel and spatial modulation. Proposal-aligned geometric convolution (SEGC) parameterizes deformable offsets in a radial–tangential frame defined on each canonical RoI. Feature-informed geometry-aware sampling (FIGAS) combines finite-difference angular sensitivity with an RoI-derived attenuation gate to adapt positive-sample selection and classification weighting during training. We compare CSGC with an Oriented R-CNN baseline under a controlled protocol that fixes the LSKNet-S backbone, data splits, training schedule, and base loss terms. On the official DOTA-v1.0 and DOTA-v2.0 Task 1 test servers, CSGC achieves mean average precision (mAP) at an intersection-over-union (IoU) threshold of 0.5, reported as $\mathbf{AP_{50}}$, of 77.58% and 53.18%. These exceed the shared baselines by 2.42 and 2.80 percentage points, respectively. Component analyses are evaluated on the validation splits. Corresponding gains on the DIOR-R and HRSC2016 test sets are 2.20 and 2.40 points, respectively. When mAP is averaged over IoU thresholds from 0.5 to 0.95, the HRSC2016 margin reaches 4.35 points. These results support the effectiveness of cross-stage geometric conditioning within the evaluated architectures and settings, with the largest observed margin under stricter localization criteria.**

***Index Terms*—Cross-stage conditioning, label assignment, oriented object detection, remote sensing, rotated RoI sampling.**

I. Introduction

ACCURATE localization of ships, aircraft, vehicles, and infrastructure is central to object-level interpretation of high-resolution aerial imagery [1], [2]. This task is difficult because targets are often densely packed and vary widely in orientation, scale, and aspect ratio. For elongated objects, axis-aligned boxes include substantial background; in crowded scenes, their overlap also makes adjacent instances difficult to separate. Oriented boxes are therefore widely used to encode both target extent and direction [3], [4]. Specifying the output geometry, however, does not determine how a detector should use orientation during feature construction, RoI sampling, and training assignment.

Geometry has consequently been incorporated into several detector interfaces, with each mechanism targeting a specific prediction, feature-alignment, sampling, or assignment problem. For box prediction, CSL [5] encodes angular periodicity, whereas GWD [6] represents rotated boxes as Gaussian distributions for regression. ReDet [7] couples an equivariant backbone with rotation-invariant RoI processing, while S2A-Net [8] aligns features with refined anchors. ARC [9] adapts convolution kernels to local orientation; EARL [10] and DAL [11] instead modify sample assignment using geometric criteria. Taken together, these developments motivate a controlled analysis of rotation-aggregated pyramid modulation, proposal-aligned RoI sampling, and feature-informed geometry-aware assignment within a common two-stage pipeline.

The central difficulty lies at the interfaces between detector stages rather than in the rotated-box representation itself. The baseline feature pyramid does not explicitly aggregate responses across canonical rotations and pyramid levels. In the RoI head, Cartesian deformable offsets are expressed in image axes rather than in directions defined by the proposal. During assignment, a fixed overlap rule treats equal IoUs alike, although the same angular error can affect elongated and compact boxes differently. Addressing these mismatches jointly offers a principled route to preserve target-aligned evidence while accounting for shape-dependent angular sensitivity during training. Figure 1 maps the three interfaces onto the detector pathway and the training-only assignment branch.

Cross-stage geometric conditioning (CSGC) addresses these interfaces through separate geometry-aware operations in the feature pyramid, RoI head, and training assignment. Cross-scale rotation-conditioned channel–spatial coupling (CSSC) forms a descriptor by aggregating rotated responses across pyramid levels, then uses it to modulate channel and spatial features. Proposal-aligned geometric convolution (SEGC) parameterizes deformable RoI-head offsets in a radial–tangential frame aligned with each proposal. Feature-informed geometry-aware sampling (FIGAS) combines finite-difference angular

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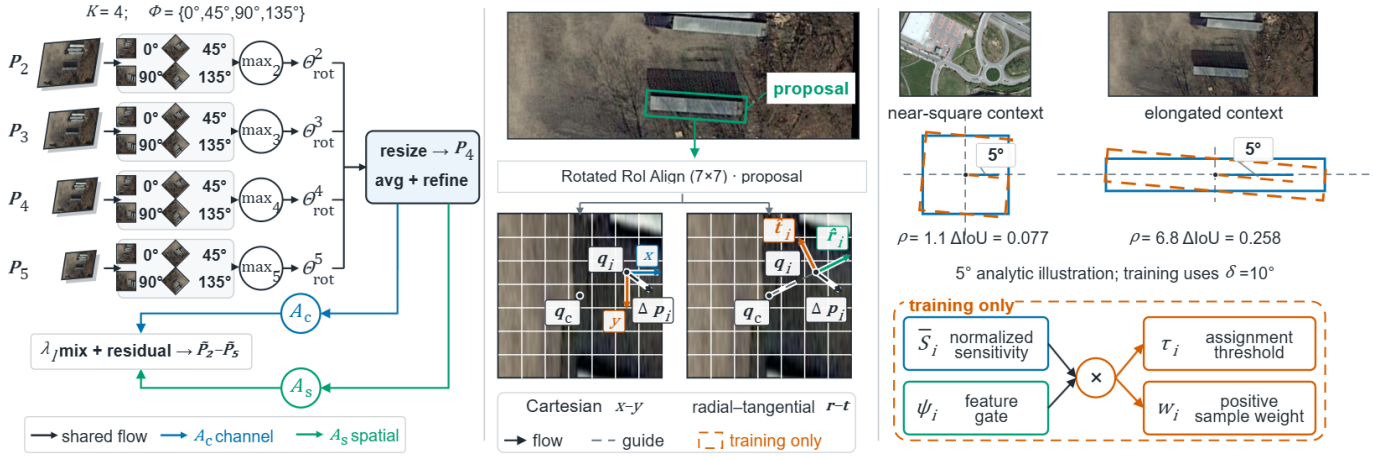


Fig. 1. Geometric conditioning at three interfaces of a two-stage oriented detector. *Left*: CSSC aggregates rotated pyramid responses across scales to produce the channel and spatial modulators A_c and A_s. *Center*: SEGC expresses RoI-head offsets in a proposal-aligned radial–tangential frame. *Right*: FIGAS combines normalized angular sensitivity $\bar{S}_i$ with the RoI-feature attenuation gate ψ_i to set the assignment threshold τ_i and positive-sample weight w_i during training. The 5° rectangles illustrate concentric self-IoU; training uses $\delta = 10^\circ$.

sensitivity with an attenuation gate computed from the detached RoI feature to adapt positive selection and classification weighting. The term *cross-stage* denotes the coordinated placement of these stage-specific operations rather than a single equivariant system. Equivariance is used only for the measured RoI-level diagnostic and is not claimed as an end-to-end property of the detector.

The principal contributions are summarized as follows:

(1) We introduce CSGC, a geometric conditioning design for two-stage oriented detection. A shared scalar derived from angular IoU sensitivity and conditioned RoI features jointly controls positive-assignment thresholds and classification weights.

(2) We develop CSSC for rotation-aggregated pyramid modulation, SEGC for radial–tangential RoI sampling, and FIGAS for feature-informed assignment and weighting.

(3) We evaluate the resulting detector on four benchmarks through matched-pipeline comparisons, component ablations, and targeted controls. Official DOTA test evaluations and a CSSC–FIGAS transfer to ReDet extend the empirical evaluation beyond the validation ablations.

II. Related Work

A. Oriented-Box Representation and Localization

Oriented-box localization introduces periodic and representation-dependent ambiguities that do not arise for axis-aligned boxes [12]. CSL and PSC recast angle estimation as circular classification or phase coding, while continuous encodings avoid discrete angular bins [4], [5], [13]. Distributional formulations map rotated boxes to Gaussian geometry and optimize Wasserstein, Kullback–Leibler, or Kalman-filter-inspired objectives [6], [14], [15]. Subsequent studies address boundary discontinuity, restricted state–angle representations, corner-sensitive overlap, and comprehensive Gaussian box models [16]–[19]. A unified transfer strategy adapts regression losses from horizontal to oriented detection using the smallest axis-aligned enclosing rectangle and two offsets [20]. While these studies primarily address box representation and localization objectives, CSGC uses geometric information to modulate pyramid features and RoI sampling and to adapt positive-assignment thresholds and classification weights.

B. Rotation-Aware Features and Aligned Sampling

Rotation-aware feature representation has developed along two broad equivariant and adaptive-filtering lines. Group-equivariant CNNs and $E(2)$-steerable networks provide formal mechanisms for transforming feature responses under rotations [21]–[23]. ORN rotates convolutional filters, whereas ReDet combines an equivariant backbone with rotation-invariant RoI features [7], [24]. FRED extends rotation-aware processing across the detector, while ARC predicts input-dependent kernel orientations [9], [25]. Large selective kernels and poly-kernel modules enlarge contextual support without imposing formal rotation equivariance [26]–[28]. The feature-side question addressed by CSSC is narrower: whether pyramid modulation benefits from descriptors aggregated across canonical rotations and feature levels.

Multiscale context supports proposal classification, while task-consistent focal loss couples classification and regression by weighting each loss according to the difficulty of the other task [29]. Pointwise frequency attention encodes local orientation cues, while coarse-to-fine supervision at both image and instance levels improves discrimination among similar subcategories [30].

Alignment methods address a distinct mismatch between sampling locations and oriented instances. RRPN generates rotated proposals, while RoI Transformer learns the transformation from horizontal proposals to rotated RoIs [31], [32]. R^3Det and S2A-Net resample features according to refined boxes or anchors in single-stage detectors [8], [33]. Oriented RepPoints represents an instance with an adaptive point set and evaluates its quality during assignment [34]. Deformable convolutions provide the general mechanism for learning non-Cartesian sampling locations [35], [36]. SEGC belongs to this

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alignment lineage but expresses RoI-head offsets in proposal-aligned radial–tangential coordinates.

C. Geometry-Aware Sample Assignment

Sample assignment determines which candidates provide classification and localization supervision. OHEM established loss-driven hard-example selection, whereas PAA models candidate quality from classification and localization losses [37], [38]. ATSS adapts the IoU threshold to candidate statistics, and OTA formulates matching as an optimal-transport problem [39], [40]. TOOD ranks and weights candidates through a task-alignment metric derived from classification and localization predictions [41]. These methods replace fixed assignment rules with statistics or signals produced during training.

Oriented detection further introduces shape and direction into the assignment criterion. DAL evaluates candidate quality using the regressed box, whereas EARL constructs an annotation-derived elliptical distribution [10], [11]. GGHL combines a Gaussian spatial prior with a regression-loss weight, and Oriented RepPoints incorporates point-set quality and feature similarity [34], [42]. Feature- or prediction-conditioned assignment is therefore well established in prior object-detection research. FIGAS instead couples an analytical estimate of angular IoU sensitivity with an attenuation gate computed from the detached RoI feature. Its scope is this joint threshold-and-weighting rule, not the general introduction of learned signals into assignment.

Recent semi-supervised oriented detectors also examine the selection and use of training supervision. ALS Teacher studies active label selection [43], while Global Focal Learning addresses learning in semi-supervised oriented detection [44]. Their setting differs from the fully supervised, candidate-level coupling of assignment thresholds and classification weights in CSGC.

D. Geometric Conditioning Across Detector Stages

Table I maps representative methods to the detector interface at which geometric information enters. ReDet spans feature extraction and RoI processing, while S2A-Net and Oriented RepPoints align sampling with instance geometry. ARC modifies feature extraction, whereas DAL, EARL, and GGHL operate through assignment. The final column distinguishes annotation-only rules from assignments that consume predictions or learned features. Coverage across more interfaces is not evidence of greater effectiveness, and the table does not compare detection accuracy or establish a causal benefit.

Within this taxonomy, CSGC applies geometric conditioning to feature-pyramid modulation, proposal-aligned RoI sampling, and training supervision. TOOD also uses a shared task-alignment metric for assignment and loss weighting [41]; FIGAS instead combines an angular IoU sensitivity estimate with a gate derived from geometrically conditioned RoI features. Specifically, FIGAS forms $c_i = \bar{S}_i \psi_i$, where $\bar{S}_i$ is the normalized angular IoU sensitivity estimate and ψ_i is a learned attenuation gate computed from the RoI features conditioned by CSSC and SEGC. The same c_i adjusts both the positive-assignment IoU threshold and the positive-sample classification weight. The feature-copy control in Table X tests the gate feature source while retaining the full detection pathway.

III. Methodology

A. Overall Framework and Notation

CSGC adopts Oriented R-CNN [45] with an LSKNet-S backbone [26], [28]. Given an aerial image $I \in \mathbb{R}^{3\times H\times W}$, an FPN [46] transforms the backbone features $\{C_l\}_{l\in\mathcal{L}}$ into the multiscale pyramid $\{P_l\}_{l\in\mathcal{L}}$, where $\mathcal{L} = \{2, 3, 4, 5\}$. The notation $F_l := P_l \in \mathbb{R}^{C\times H_l\times W_l}$ is used with $C = 256$. The oriented proposal head uses anchor aspect ratios $\{0.5, 1.0, 2.0\}$ at every pyramid level, and rotated RoI Align maps each proposal to a feature tensor $r_i \in \mathbb{R}^{C\times 7\times 7}$.

An oriented box follows the MMRotate long-edge convention (le90), $b = (x, y, w, h, \theta)$, with $w \geq h > 0$ and $\theta \in [-90^\circ, 90^\circ)$. Its aspect ratio is $\rho = w/h \geq 1$. The operator R_ϕ rotates a feature map spatially about its center by ϕ, using bilinear resampling and zero padding; $\odot$ denotes elementwise multiplication, $[\cdot\,;\,\cdot]$ channelwise concatenation, and $\sigma(\cdot)$ the logistic sigmoid. Additional symbols are introduced at their first use.

Figure 2 follows the detector data flow and locates the three conditioning interfaces. CSSC acts on the FPN before instance hypotheses are formed, and hence requires no object-specific orientation. SEGC uses the geometry of each oriented proposal to parameterize RoI-head sampling. FIGAS is confined to training and combines candidate–ground-truth geometry with an attenuation coefficient derived from the corresponding RoI representation. The components consequently exchange information through the detector pathway rather than through a separate shared tensor.

B. Cross-Scale Rotation-Conditioned Channel–Spatial Coupling (CSSC)

Conventional channel and spatial attention modules, including SE-Net [47], CBAM [48], ECA [49], coordinate attention [50], and PKINet [27], recalibrate feature responses without explicitly aggregating their rotated counterparts. CSSC instead derives both gates from a descriptor shared across orientations and pyramid levels. The qualifier *channel–spatial* refers exclusively to the two modulation gates and does not denote an auxiliary semantic-supervision branch.

Let $\Phi = \{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$ be the set of $K = 4$ canonical rotations. For each level $l \in \mathcal{L}$, a pixelwise maximum on the common output grid forms the rotation-conditioned response envelope

$$\Theta^l_{\mathrm{rot}} = \max_{\phi\in\Phi} R_\phi(F_l). \tag{1}$$

The maximum is taken after rotation, without inverse alignment. Consequently, Θ^l_{rot} is a rotation-augmented response envelope rather than an orientation-preserving or formally rotation-invariant representation. Under le90, orientation is periodic over 180°, and the selected angles sample the orientation interval at 45° increments. Resampling cost grows linearly with the number of rotated copies K.

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TABLE I
DESIGN-SCOPE COMPARISON WITH REPRESENTATIVE ORIENTED-DETECTION METHODS. THE TABLE CHARACTERIZES WHERE EACH METHOD INJECTS GEOMETRIC STRUCTURE.

Method	Feature geometry	Sampling geometry	Assignment geometry	Pred./feature → assign.
ORN [24]	✓(rotated filters)	No	No	No
ReDet [7]	✓(strict equiv.)	✓(RiRoI)	No	No
S2A-Net [8]	✓(FAM)	✓(AlignConv)	No	No
ARC [9]	✓(rotated kernels)	No	No	No
DAL [11]	No	No	✓(matching degree)	partial (regressed box)
EARL [10]	No	No	✓(elliptical dist.)	No
GGHL [42]	No	No	✓(Gaussian heatmap)	partial (regression loss)
O. RepPoints [34]	✓(point features)	✓(adaptive points)	✓(quality measure)	✓(point-wise features)
CSGC	✓(CSSC)	✓(SEGC)	✓(FIGAS)	✓(ψ_i)

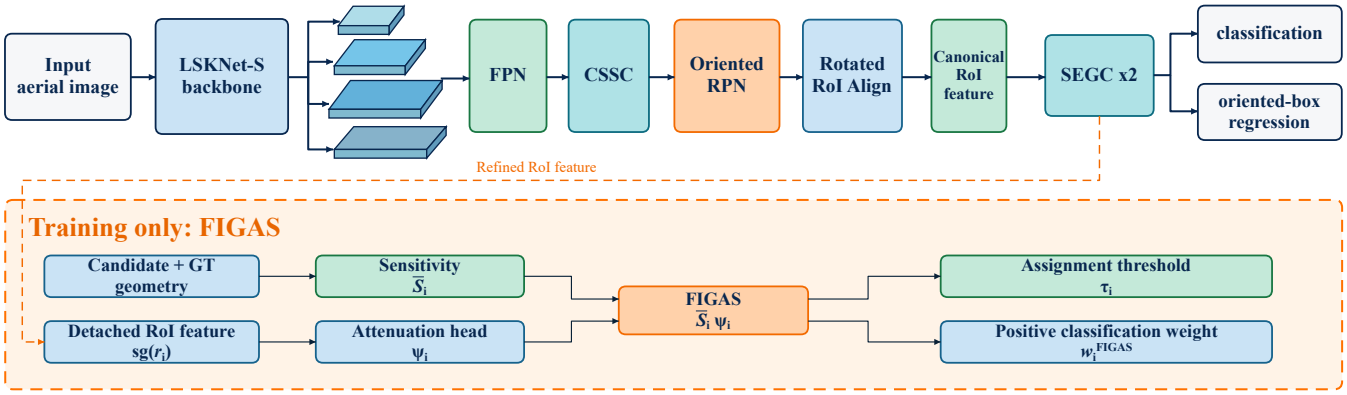


Fig. 2. Overall architecture of CSGC. The inference pathway uses CSSC, proposal generation, rotated RoI Align, two SEGC layers, and classification/regression heads. The training-only FIGAS branch receives box geometry and the detached RoI representation as parallel inputs. Its shared conditioning product determines both the positive-assignment threshold and the positive-sample classification weight.

The per-level envelopes are aligned to the stride-16 grid of P_4 and averaged before either gate is computed. Denoting bilinear resizing from level l to the P_4 reference grid by $\mathcal{B}_{l\to 4}$, the cross-scale descriptor is

$$D = \frac{1}{|\mathcal{L}|}\sum_{l\in\mathcal{L}} \mathcal{B}_{l\to 4}\big(\Theta^{l}_{\mathrm{rot}}\big), \qquad D \in \mathbb{R}^{256\times H/16\times W/16}. \quad (2)$$

A shared two-layer convolutional block refines D once:

$$\hat{D} = \mathrm{Conv}^{180\to 256}_{3\times 3}\big(\mathrm{ReLU}\big(\mathrm{Conv}^{256\to 180}_{3\times 3}(D)\big)\big). \quad (3)$$

Cross-scale coupling precedes gate generation: every gate is conditioned on the aggregate D, rather than on one pyramid level in isolation.

The refined descriptor produces a channel gate $A_c \in (0,1)^{C\times 1\times 1}$ and a spatial gate $A_s \in (0,1)^{1\times H/16\times W/16}$:

$$A_c = \sigma\Big(\mathrm{MLP}(\mathrm{GAP}(\hat{D}))\Big), \quad (4)$$

$$A_s = \sigma\Big(\mathrm{Conv}_{7\times 7}\Big([\,\mathrm{AvgPool}_c(\hat{D});\ \mathrm{MaxPool}_c(\hat{D})\,]\Big)\Big), \quad (5)$$

where GAP averages over the two spatial dimensions, while $\mathrm{AvgPool}_c$ and $\mathrm{MaxPool}_c$ operate across channels. The MLP is a $256 \to 16 \to 256$ bottleneck with an intermediate ReLU and no normalization. The sigmoid bounds the two modulation coefficients but does not, because of the residual connection below, constrain the final feature magnitude to decrease.

For level l, A_s is bilinearly resized to $H_l \times W_l$, denoted by $A_s^{\uparrow l}$, whereas A_c is broadcast spatially. A trainable logit g_l, initialized to zero, defines the level-specific mixture coefficient $\lambda_l = \sigma(g_l)$ and therefore $\lambda_l = 0.5$ at initialization. The conditioned output is

$$\tilde{F}_l = F_l + F_l \odot \big(\lambda_l\, A_c + (1-\lambda_l)\, A_s^{\uparrow l}\big), \quad (6)$$

so the two gates contribute equally at initialization. Both gates are computed once from $\hat{D}$; level specificity enters only through g_l and the resizing of A_s. The unrotated feature F_l remains on the identity path, and rotated copies are restricted to the conditioning branch. Figure 3 summarizes the module data flow.

C. Proposal-Aligned Geometric Convolution (SEGC)

Standard and deformable convolutions do not require RoI-head offsets to follow the orientation of a proposal. SEGC instead expresses every learned displacement in a radial–tangential basis defined on the canonical RoI grid. Proposal anisotropy enters through the proposal-to-RoI transformation, whereas the basis depends only on canonical-grid coordinates. The resulting parameterization is an invertible change of

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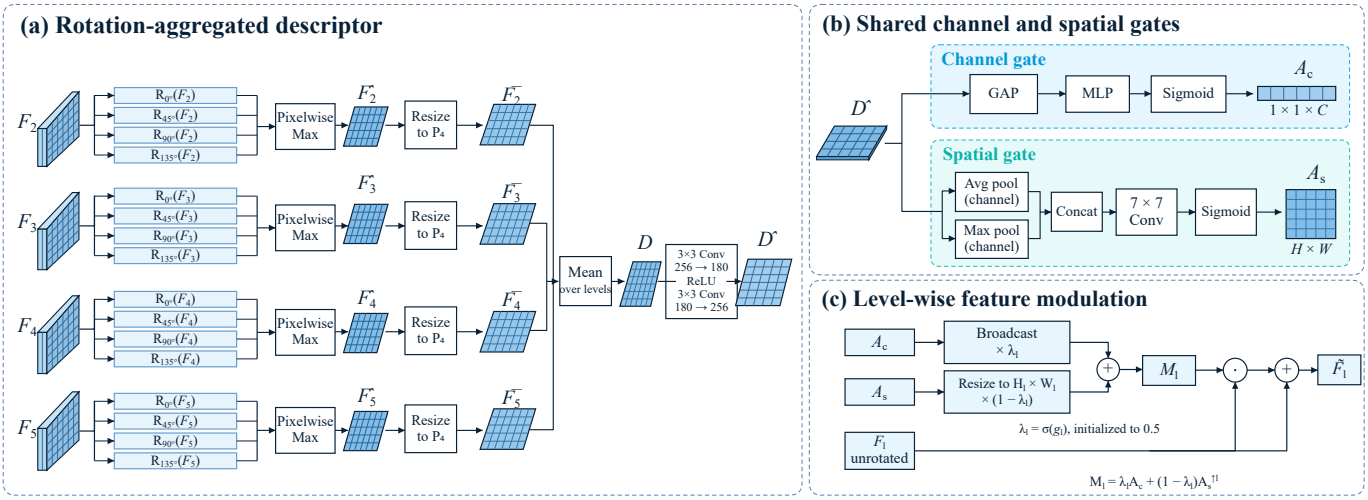


Fig. 3. Cross-scale rotation-conditioned channel–spatial coupling. Parallel rotated copies of each F_l enter a pixelwise maximum, are resized to the common P_4 grid and averaged, then pass through two shared 3×3 refinement layers. Parallel channel and spatial gates form $\lambda_l A_c + (1-\lambda_l)A_s^{\uparrow l}$; multiplication by the unrotated F_l and residual addition produce $\tilde{F}_l$.

coordinates for two-dimensional offsets and therefore supplies an inductive bias without restricting the unconstrained offset function class or guaranteeing equivariance.

Let $\mathbf{x} : \mathbb{R}^2 \to \mathbb{R}^C$ denote the bilinearly sampled input feature field on the canonical 7×7 grid, and let $\mathbf{y}$ denote the output field. For output location $\mathbf{q}_0$, grid center $\mathbf{q}_c$, and the ith regular offset $\mathbf{p}_i$ of a 3×3 kernel, define $\mathbf{u}_i = \mathbf{q}_0 + \mathbf{p}_i - \mathbf{q}_c$, with $i \in \{1, \ldots, N\}$ and $N = 9$. For $\mathbf{u}_i \neq \mathbf{0}$, the orthonormal basis is

$$\hat{\mathbf{r}}_i = \frac{\mathbf{u}_i}{\|\mathbf{u}_i\|_2}, \qquad \hat{\mathbf{t}}_i = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \hat{\mathbf{r}}_i, \tag{7}$$

where the central location uses $\hat{\mathbf{r}}_i = (1, 0)^\top$ and $\hat{\mathbf{t}}_i = (0, 1)^\top$. Rotated RoI Align makes the canonical grid proposal-aligned in image coordinates; proposal alignment alone does not make the learned operator rotation-equivariant.

An offset branch predicts the scalar pair $(\delta r_i, \delta t_i)$ for every kernel location, yielding $2N$ output channels. A parallel branch predicts a logit a_i and modulation coefficient $m_i = \sigma(a_i) \in (0, 1)$. The Cartesian displacement recovered from the proposal-aligned coordinates is

$$\Delta \mathbf{p}_i = \delta r_i \, \hat{\mathbf{r}}_i + \delta t_i \, \hat{\mathbf{t}}_i, \tag{8}$$

and deformable aggregation becomes

$$\mathbf{y}(\mathbf{q}_0) = \sum_{i=1}^{N} m_i \, \mathbf{W}_i \, \mathbf{x}(\mathbf{q}_0 + \mathbf{p}_i + \Delta \mathbf{p}_i), \tag{9}$$

where $\mathbf{W}_i$ is the channel-mixing kernel weight at location i; fractional coordinates use bilinear interpolation with zero padding. SEGC replaces both consecutive 3×3 RoI-head convolutions. Each offset branch maps $C = 256$ channels to $2N = 18$ channels with a 3×3 convolution, unit stride, unit padding, and no normalization. Zero-initialized offset weights and biases recover the regular sampling grid at initialization. Each modulation branch maps 256 channels to $N = 9$ logits with a parallel 3×3 convolution followed by a sigmoid. Gradients through fractional coordinates follow standard deformable convolution.

The proposal-aligned basis changes the learning coordinates rather than enforcing a transformation law on the predicted scalars $(\delta r_i, \delta t_i)$. Rotational stability is therefore evaluated with the RoI-level diagnostic

$$E_{\text{equiv}}(\phi) = \frac{\| f_{\text{conv}}(R_\phi(\mathbf{x})) - R_\phi(f_{\text{conv}}(\mathbf{x})) \|_2}{\max(\| f_{\text{conv}}(\mathbf{x}) \|_2, \varepsilon)}, \tag{10}$$

where f_{conv} denotes the RoI operator under evaluation, $\| \cdot \|_2$ covers all spatial and channel entries, and $\varepsilon = 10^{-5}$ prevents division by zero. Evaluation uses 1,000 matched proposals from a fixed image set at $\phi \in \{0°, 45°, \ldots, 315°\}$. Proposal coordinates are transformed consistently between each original–rotated image pair. Figure 4 summarizes the operator and its proposal-aligned coordinate frame. The diagnostic measures feature-space consistency rather than detection accuracy; Sec. IV-E reports the corresponding measurements.

The operator rows use the corresponding trained configurations in Tables XI and IX. They share the fixed proposal set, RoI measurement interface, eight rotations, bilinear resampling, and zero-padding convention. Operator comparisons use the same proposal transformations rather than independently selected detection outputs. The Identity row is defined separately as the strict operator $f(\mathbf{x}) = \mathbf{x}$, for which Eq. (10) is zero.

D. Feature-Informed Geometry-Aware Sampling (FIGAS)

Oriented assigners differ in their reliance on model state (Sec. II-C). EARL [10] derives its rule from annotated geometry, whereas DAL [11] and GGHL [42] also use prediction-dependent quantities. FIGAS combines an analytical angular-sensitivity factor from box geometry with a learned attenuation coefficient from the RoI representation. The term *sampling* denotes training-sample selection and weighting, rather than the spatial sampling offsets introduced by SEGC.

Angular sensitivity. The baseline assigner first establishes a provisional association between candidate $b_i = (x_i, y_i, w_i, h_i, \theta_i)$ and ground truth g_i. Let b_i^+ and b_i^- retain

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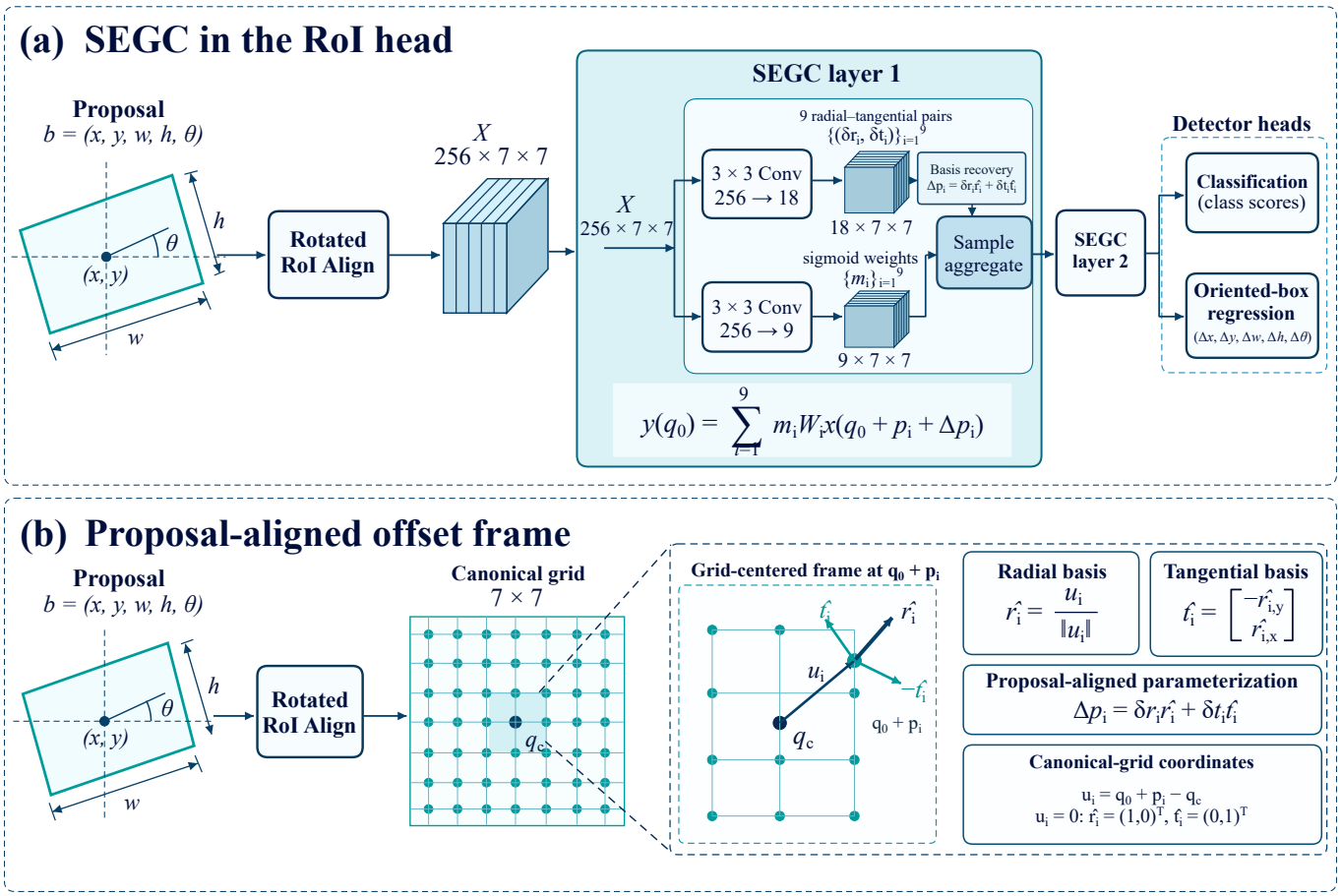


Fig. 4. Proposal-aligned geometric convolution. Rotated RoI Align supplies a canonical 7×7 grid. Each of two layers predicts $(\delta r_i, \delta t_i)$ and modulation coefficients, with $\mathbf{u}_i = \mathbf{q}_0 + \mathbf{p}_i - \mathbf{q}_c$ defining the radial–tangential basis. The recovered displacement is $\Delta\mathbf{p}_i = \delta r_i \hat{\mathbf{r}}_i + \delta t_i \hat{\mathbf{t}}_i$. The basis is an inductive bias rather than a guarantee of equivariance.

(x_i, y_i, w_i, h_i) while replacing the angle by $\theta_i + \delta$ and $\theta_i - \delta$, respectively. Each perturbed box is mapped back to le90 while preserving its physical rectangle and the long-edge condition $w \geq h$. With angular quantities measured in radians, the finite-difference sensitivity is

$$S_i = \frac{1}{\sqrt{2}\,\delta}\left(\left[\Delta_i^+\right]^2 + \left[\Delta_i^-\right]^2\right)^{1/2}, \tag{11}$$

where $\Delta_i^{\pm} = \mathrm{IoU}(b_i^{\pm}, g_i) - \mathrm{IoU}(b_i, g_i)$ and IoU is evaluated between quadrilateral polygons. The default perturbation is $\delta = \pi/18$ (10°); the sensitivity analysis explicitly varies it. Equation (11) is the root-mean-square of the two IoU changes divided by δ, equivalently a denominator of $\sqrt{2}\,\delta$, with S_i in rad^{-1}. At differentiable orientations, S_i approximates the magnitude of the angular derivative; the symmetric two-sided construction also avoids choosing one side of an isolated le90 boundary. A finite perturbation sacrifices infinitesimal locality in exchange for numerical stability of polygon-IoU evaluation. The value of δ is selected on DOTA-v2.0 and retained for every other benchmark; Sec. IV-D4 reports the associated sensitivity analysis.

A small-angle calculation provides a local geometric motivation for assigning larger angular-sensitivity weights to elongated boxes. For two concentric, equal-size rectangles with aspect ratio $\rho = w/h \geq 1$, the intersection satisfies $A_\cap(\theta) \approx wh - \frac{1}{4}(w^2+h^2)|\theta|$, and since the union is $2wh - A_\cap$, it grows by the same first-order amount. Hence

$$\mathrm{IoU}(\theta) \approx \frac{wh - \frac{1}{4}(w^2+h^2)|\theta|}{wh + \frac{1}{4}(w^2+h^2)|\theta|},$$
$$\left|\frac{\partial\,\mathrm{IoU}}{\partial\theta}\right|_{\theta=0^\pm} = \frac{w^2+h^2}{2wh} = \tfrac{1}{2}\left(\rho + \rho^{-1}\right),$$

whose leading term is $\rho/2$ for elongated boxes. Evaluating the exact polygon IoU under a 5° perturbation at $\rho = 1.1$, 3.2, and 6.8 gives drops of 0.077, 0.136, and 0.258, so the first-order expression overstates the finite-angle effect by 12–18% while preserving the ordering the weighting uses (Fig. 5a). The calculation concerns concentric self-IoU and does not by itself predict candidate-to-ground-truth behavior; its aspect-ratio implication is treated as a working hypothesis and examined empirically in Sec. IV-E. The score is normalized as $\bar{S}_i = \min(S_i, S_{\max})/S_{\max} \in [0,1]$, where $S_{\max} = \frac{1}{2}(\rho_{\max} + \rho_{\max}^{-1})$ is the analytical sensitivity evaluated at the largest annotated aspect ratio $\rho_{\max}$ of the training split.

Feature-derived attenuation coefficient. A compact head maps the RoI tensor r_i to a scalar logit z_i through global average pooling, a $256 \to 16$ linear layer, layer normalization, ReLU, and a $16 \to 1$ linear layer. The coefficient

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is $\psi_i = \sigma(z_i) \in (0, 1)$. The head contains 4,161 trainable parameters and is used only during training. Its input is the SEGC output when SEGC is enabled and the standard RoI-head feature otherwise, preserving a common definition across all ablation cells.

The head receives $\mathrm{sg}(r_i)$, where sg is the identity in the forward pass and has zero derivative in the backward pass. Detachment prevents the attenuation objective from modifying the shared RoI representation, while gradients continue to update the attenuation-head parameters. No auxiliary target supervises ψ_i. For a fixed positive set and nonnegative per-sample classification loss $\ell_{\mathrm{cls},i}$, the weighting path yields

$$\frac{\partial \mathcal{L}_{\mathrm{cls}}^{\mathrm{FIGAS}}}{\partial z_i} = \frac{\beta\, \bar{S}_i\, \ell_{\mathrm{cls},i}}{\max(1, |\mathcal{P}|)}\, \psi_i(1 - \psi_i) \geq 0, \tag{12}$$

where $\mathcal{P}$ is the positive set. The thresholding operation is a nondifferentiable forward assignment rule and contributes no gradient to z_i; attenuation-head learning therefore proceeds only through Eq. (12). Gradient descent tends to reduce ψ_i more strongly for samples with larger weighted classification loss. No interior stationary point follows from the weighting path alone, although sigmoid saturation slows the drift. Accordingly, ψ_i is treated as a schedule-dependent attenuation coefficient rather than a calibrated measure of feature quality, confidence, or sample difficulty.

Assignment and weighting. Both factors enter the assigner through a relaxed positive threshold and a positive-sample weight:

$$\tau_i = \tau_0 \left(1 - \alpha\, \bar{S}_i\, \psi_i\right), \qquad w_i^{\mathrm{FIGAS}} = 1 + \beta\, \bar{S}_i\, \psi_i, \tag{13}$$

with base threshold $\tau_0 = 0.6$. Since $\bar{S}_i, \psi_i \in [0, 1]$, the default $\alpha = 0.3$ confines τ_i to $[0.42, 0.60]$ without clipping, while $w_i^{\mathrm{FIGAS}} \in [1, 1 + \beta]$. A candidate receives the relaxed threshold only when both angular sensitivity and feature-derived attenuation are high. The baseline treatment of negatives and localization targets remains unchanged. The defaults $\alpha = 0.3$ and $\beta = 0.5$ are selected on DOTA-v2.0 validation and retained elsewhere, including the frozen official submissions; Sec. IV-D4 examines both hyperparameters. Figure 5 traces the analytical and learned factors to their two training-time outputs.

Let $c_i = \bar{S}_i \psi_i$ denote the shared conditioning scalar in Eq. (13). For $\alpha > 0$, eliminating c_i yields

$$w_i^{\mathrm{FIGAS}} - 1 = \frac{\beta}{\alpha}\left(1 - \frac{\tau_i}{\tau_0}\right).$$

For fixed positive α and β, a larger c_i lowers the assignment threshold and increases the positive-sample classification weight. When $\alpha = 0$, $\tau_i = \tau_0$ and $w_i^{\mathrm{FIGAS}} = 1 + \beta c_i$.

E. Training Objective and Computational Scope

Let $\mathcal{P}$ and $\mathcal{N}$ denote the positive and negative candidate sets produced by the assignment rule. Positive focal losses are reweighted, whereas negative focal losses retain unit weight:

$$\mathcal{L}_{\mathrm{cls}}^{\mathrm{FIGAS}} = \frac{1}{\max(1, |\mathcal{P}|)} \left(\sum_{i \in \mathcal{P}} w_i^{\mathrm{FIGAS}} \ell_{\mathrm{cls},i} + \sum_{j \in \mathcal{N}} \ell_{\mathrm{cls},j} \right). \tag{14}$$

The complete training objective combines classification, localization, and periodic angle terms:

$$\mathcal{L} = \mu_1\, \mathcal{L}_{\mathrm{cls}}^{\mathrm{FIGAS}} + \mu_2\, \mathcal{L}_{\mathrm{loc}}^{\mathrm{rIoU}} + \mu_3\, \mathcal{L}_{\mathrm{ang}}^{\mathrm{sin}}, \tag{15}$$

where $\ell_{\mathrm{cls},i}$ is focal loss [51], $\mathcal{L}_{\mathrm{loc}}^{\mathrm{rIoU}}$ is rotated-IoU localization loss [15], and $\mathcal{L}_{\mathrm{ang}}^{\mathrm{sin}}$ denotes the shared sinusoidally modulated angle loss used to respect angular periodicity. The coefficients $(\mu_1, \mu_2, \mu_3) = (0.4, 0.2, 0.4)$ are selected on DOTA-v2.0 and retained on the other benchmarks.

All eight factorial configurations, including the baseline, use the loss families and coefficients in Eq. (15). Disabling FIGAS restores $\tau_i = \tau_0$ and $w_i^{\mathrm{FIGAS}} = 1$; the rotated-IoU localization term, periodic angle term, and coefficients (μ_1, μ_2, μ_3) remain fixed. Active CSSC and SEGC modules change the forward representation, whereas active FIGAS additionally changes positive assignment and classification weighting. No factorial cell substitutes a different loss family, preventing loss selection from becoming an unablated fourth intervention.

At inference, only CSSC and SEGC remain in the detector. The sensitivity calculation, attenuation head, adaptive threshold, and positive-sample weight are discarded together with the complete FIGAS branch. Section IV-C reports inference complexity and the separately measured training-time overhead.

IV. Experiments

The experiments examine detection accuracy, strict-overlap localization, computational cost, component contributions, and geometric behavior. Matched-pipeline comparisons and a complete 2^3 design quantify accuracy and component differences under a shared detector, while stage-local alternatives and a capacity-matched RoI-support perturbation probe competing explanations. Published results from unmatched systems provide external context only and support no controlled module-level claim.

A. Experimental Setup

Datasets and evaluation protocol. Four public optical remote-sensing benchmarks define the evaluation domain. **DOTA-v1.0** [52] contains 2,806 images in 15 categories, partitioned into 1,411 training, 458 validation, and 937 test images. **DOTA-v2.0** [1] expands the benchmark to 11,268 images and 18 categories, with additional small-object instances and a heavier category tail. **DIOR-R** [53] contains 23,463 images and 20 categories under the official 11,725/11,738 train–test partition. **HRSC2016** [54] contains 1,061 ship images under the standard 436/181/444 train–validation–test split. No benchmark is repartitioned.

DOTA images are divided into 800×800 crops with a stride of 200 pixels. Cross-crop detections are merged by rotated NMS at an IoU threshold of 0.1, and oriented boxes follow the le90 convention. The primary endpoint is mAP@0.5 under the PASCAL VOC 2007 interpolation protocol; the same VOC07 convention is used for HRSC2016.

The DOTA-v1.0 and DOTA-v2.0 matched-pipeline comparisons and component analyses are evaluated on their validation splits using the local implementation of the official devkit

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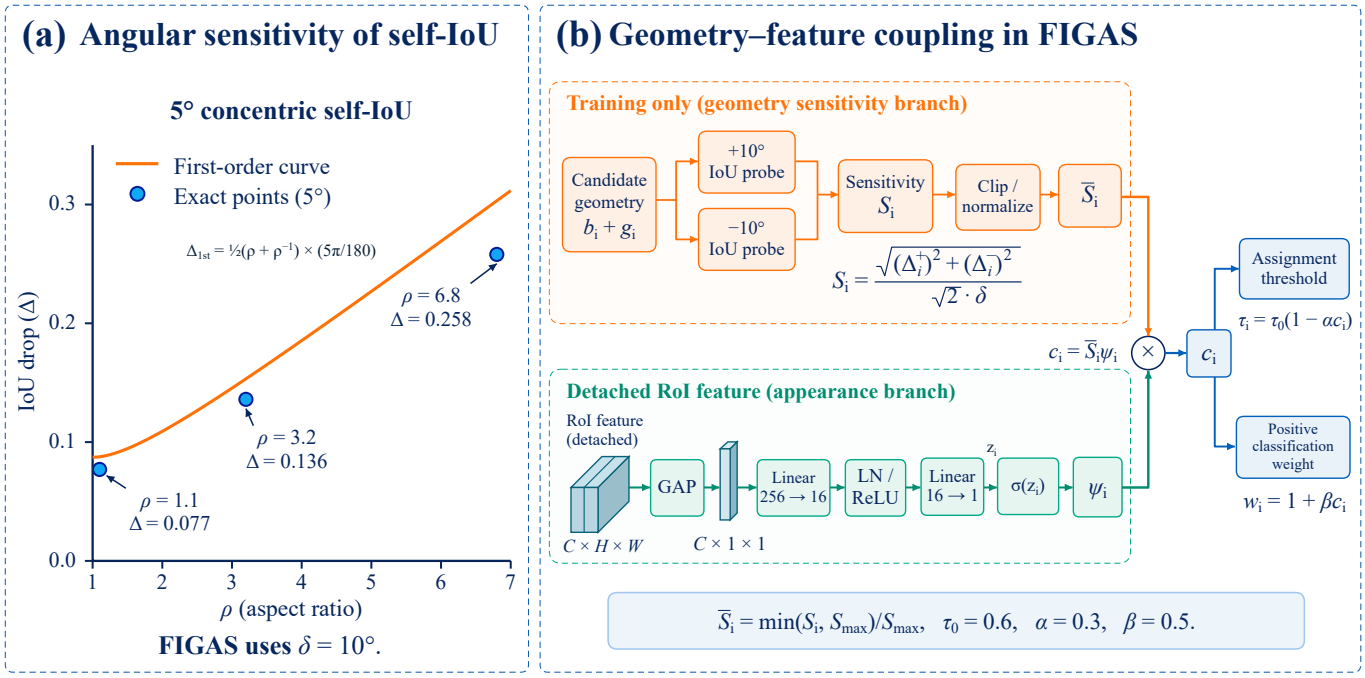


Fig. 5. Feature-informed geometry-aware sampling. The first-order self-IoU approximation and exact polygon evaluations use an illustrative 5° perturbation; the training default is 10°. Candidate–ground-truth geometry gives $\bar{S}_i$, while the attenuation head maps $\mathrm{sg}(r_i)$ through logit z_i to ψ_i. Their product $c_i = \bar{S}_i\psi_i$ controls threshold and weight in parallel. The sensitivity uses the RMS of the two IoU changes divided by δ; stop-gradient preserves attenuation-head learning.

protocol. DIOR-R and HRSC2016 results are measured on the official test splits. The HRSC2016 validation split is used only for the hyperparameter analysis in Sec. IV-D4, leaving the test split outside model selection. Parameter fitting uses the official training split of each benchmark.

The DOTA validation matrix is used for configuration selection and component analysis; these values remain development-set measurements. After selection, $\alpha = 0.30$, $\beta = 0.50$, $\tau_0 = 0.60$, $K = 4$, $\delta = \pi/18$, and $(\mu_1, \mu_2, \mu_3) = (0.4, 0.2, 0.4)$ are frozen for the official Task 1 submissions. Table III reports the shared baseline and CSGC separately on DOTA-v1.0 test and DOTA-v2.0 test-dev. Test-server estimates are kept distinct from the validation ablations and from published results obtained under other protocols.

Secondary localization analysis reports AP_{50}, AP_{75}, and mAP@0.5:0.95 on DOTA-v2.0, DIOR-R, and HRSC2016. The local COCO-style evaluator applies exact polygon IoU at ten thresholds from 0.50 to 0.95 with `maxDets` $= 100$. AP_{50} records performance at one overlap threshold, whereas the threshold average places additional weight on localization quality at stricter overlaps.

Implementation and reporting conventions. A fixed optimization recipe is used for every matched configuration within a benchmark. AdamW [55] uses an initial learning rate of 10^{-4}, weight decay of 0.05, cosine annealing, a 500-step linear warm-up, and gradient clipping at $\mathrm{max_norm} = 35$. DOTA-v1.0 and DOTA-v2.0 models are trained for 12 epochs; DIOR-R and HRSC2016 models are trained for 36 epochs.

Data augmentation comprises horizontal flipping, rotations in $\{0°, 90°, 180°, 270°\}$, multiscale resizing with factors $\{0.5, 0.75, 1.0, 1.25, 1.5\}$, mosaic stitching [56], and HSV jitter. Training uses PyTorch 2.0.1, CUDA 11.8, MMRotate 1.0.0rc1 [57], mixed precision, and two RTX 3090 GPUs with two images per GPU. Identical software, augmentation, and optimization settings are retained across the matched comparisons.

Throughput is measured with batch size 1 after 200 warm-up iterations and averaged over 500 timed iterations, using the same per-crop timing procedure for the compared configurations. FPS characterizes crop-level processing; full-scene latency additionally depends on crop generation and merging.

The matched-pipeline comparisons in Table II and the DOTA-v2.0 factorial configurations use five fixed seeds: 3407, 42, 2023, 1234, and 7777. The CBAM-L, DCNv2, and FIGAS remeasurements in Table XI use seeds 3407, 42, and 2023. Sample SD is reported for the DOTA-v2.0 baseline, full configuration, and full-model RoI-support perturbation control. Between-configuration comparisons are interpreted as differences in reported means.

B. Detection Performance

Matched-pipeline comparison. Table II evaluates whether coordinated conditioning improves the shared two-stage detector under fixed data splits, optimization, augmentation, and objective. CSGC records 78.90, 53.88, 66.80, and 90.70 mAP@0.5 on DOTA-v1.0, DOTA-v2.0, DIOR-R, and HRSC2016, respectively. The corresponding differences from the reference baseline are +3.10, +3.66, +2.20, and +2.40 points. Positive matched differences occur on all four benchmarks, while their magnitude varies across datasets.

The ReDet-inspired, EARL-inspired, and ARC-inspired configurations transplant one stage-local geometric mecha-

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nism into the common pipeline; none reproduces the cited detector end to end. The ReDet-inspired row, for example, omits the group-equivariant backbone responsible for the original method's equivariance. Among the selected transplants, the ARC-inspired configuration gives the highest DOTA-v2.0 mean at 52.02, compared with 53.88 for CSGC. The resulting 1.86-point difference is stage-combined and cannot isolate a single component or interaction; the factorial and perturbation analyses address those narrower questions.

On HRSC2016, the matched AP_{50} values occupy a comparatively narrow range from 88.30 to 90.70, and CSGC exceeds the baseline by 2.40 points. Table VI complements the single-threshold result with AP_{75} and mAP@0.5:0.95. Absolute values need not reproduce the original publications because every row in Table II is retrained under the common protocol; only within-table differences support controlled comparisons.

Official test-server evaluation. Table III provides an independent endpoint after freezing the selected hyperparameters. The AP_{50} improvements are 2.42 and 2.80 points on DOTA-v1.0 and DOTA-v2.0, respectively; AP_{75} improves by 1.95 and 2.88 points. The official test results are not substituted into the validation component tables.

Complete detectors and second-family transfer. Table IV compares complete S2A-Net and ReDet with the shared two-stage pipeline while retaining their defining architectures. Complete ReDet retains its group-equivariant backbone, distinguishing it from the stage-local transplant in Table II. The transferred conditioning configuration includes CSSC and FIGAS and raises ReDet mAP@0.5 from 51.45 to 53.55, a 2.10-point difference. Oriented R-CNN+CSGC exceeds complete S2A-Net and ReDet by 2.18 and 2.43 points. The transfer supports the tested second family, rather than a detector-independent claim.

Published results under unmatched protocols. Table V records selected results reported by recent transformer-based oriented detectors under their original evaluation settings. OrientedFormer reports 54.27 on DOTA-v2.0 with ResNet-50 and a 12-epoch schedule; its DIOR-R result is 67.28, while RO^2-DETR reports 72.26 on DIOR-R. Both DIOR-R values are higher than the matched-pipeline CSGC mean of 66.80. Backbone, training data, schedule, split, and evaluation differences prevent a direct method ranking or a module-effect inference. A frozen common-protocol evaluation would be required to determine whether the published gaps persist under matched conditions.

Localization under stricter overlap criteria. Table VI separates the single-threshold VOC07 measure from COCO AP_{50}, AP_{75}, and mAP@0.5:0.95. On DOTA-v2.0, CSGC raises COCO mAP@0.5:0.95 from 27.85 to 30.62, an absolute margin of 2.77 points and a proportional margin of 9.9%. The corresponding VOC07 mAP@0.5 margin is 3.66 points, or 7.3% relative to the baseline. Averaging over stricter overlap thresholds therefore preserves the improvement observed at IoU 0.5.

The DOTA-v2.0 AP_{75} margins are 2.15 points for SEGC alone and 4.05 points for the full configuration, compared with AP_{50} margins of 1.15 and 3.55 points, respectively. A larger SEGC margin at the stricter threshold is compatible with the lower feature-space equivariance discrepancy reported in Sec. IV-E. Neither observation establishes a causal link, nor do the aggregate AP values separate localization, recall, and classification effects.

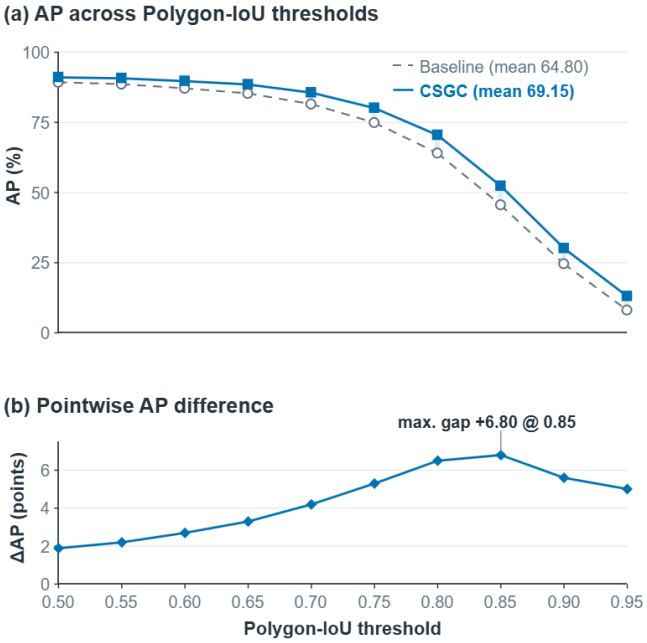


Fig. 6. Threshold-wise AP on HRSC2016. *(a)* Baseline and CSGC results at the ten COCO polygon-IoU thresholds, with mean AP values of 64.80 and 69.15. *(b)* Pointwise difference $AP_{CSGC} - AP_{baseline}$; the maximum is +6.80 points at IoU 0.85. The profile is descriptive and does not identify a specific error source.

VOC07 mAP@0.5 and COCO AP_{50} remain distinct estimates despite sharing the same nominal threshold. Relative to VOC07, COCO AP_{50} changes by -0.47, -0.45, and $+0.85$ points on DOTA-v2.0, DIOR-R, and HRSC2016, respectively. The COCO evaluator caps detections at `maxDets` $= 100$ per image; dense crops can therefore lose part of the high-recall tail, whereas the cap is less restrictive for images containing fewer objects. Interpolation rules provide an additional source of variation, and difficult or ignored instances are excluded under both conventions. Cross-protocol differences consequently reflect evaluation design as well as detector output.

The HRSC2016 mAP@0.5:0.95 margin reaches 4.35 points (64.80 $\rightarrow$ 69.15), compared with 2.40 points under VOC07 mAP@0.5. Elongated ships are more sensitive to angular misalignment, making the larger strict-overlap margin geometrically plausible; the cross-benchmark association remains descriptive rather than mechanistic. Figure 6 resolves the HRSC2016 result by threshold: the pointwise difference peaks at +6.80 points at IoU 0.85 and narrows toward both endpoints. The corresponding DIOR-R margin is 2.38 points.

C. Computational Cost and Accuracy–Throughput Trade-offs

Table VII reports inference cost under a common implementation and hardware setting. The full configuration contains 31.93 M parameters, a 0.91 M (2.93%) increase over the

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TABLE II
MATCHED-PIPELINE DETECTION ACCURACY ON FOUR BENCHMARKS. RESULTS ARE FIVE-RUN MEAN MAP@0.5 (%; HIGHER IS BETTER). EVERY CONFIGURATION USES LSKNET-S AND THE PROTOCOL OF SEC. IV-A. REDET-, EARL-, AND ARC-INSPIRED VARIANTS TRANSPLANT A STAGE-LOCAL GEOMETRIC MECHANISM INTO THE COMMON PIPELINE. BOLD MARKS THE LARGEST MEAN IN EACH COLUMN.

Method	DOTA-v1.0	DOTA-v2.0	DIOR-R	HRSC2016
Oriented R-CNN baseline [45]	75.80	50.22	64.60	88.30
ReDet-inspired variant [7]	76.95	51.45	65.12	88.92
EARL-inspired variant [10]	77.14	51.88	65.36	89.15
ARC-inspired variant [9]	77.46	52.02	65.41	89.24
CSGC	**78.90**	**53.88**	**66.80**	**90.70**

TABLE III
OFFICIAL DOTA TASK 1 TEST-SERVER RESULTS (%). IMPROVEMENTS ARE CALCULATED FROM THE UNROUNDED OUTPUTS. VOC MAP EQUALS AP_{50} AND IS REPORTED ONCE. RESULTS ARE SEPARATE FROM THE FIVE-SEED VALIDATION MEANS.

Configuration	AP_{50}	AP_{75}	mAP@0.5:0.95
DOTA-v1.0 test			
Baseline	75.16	47.57	45.31
CSGC	77.58	49.52	46.58
Δ	+2.42	+1.95	+1.27
DOTA-v2.0 test-dev			
Baseline	50.38	27.17	27.78
CSGC	53.18	30.04	30.29
Δ	+2.80	+2.88	+2.50

TABLE IV
COMPLETE ORIENTED DETECTORS ON DOTA-V2.0 VALIDATION UNDER SHARED DATA, TRAINING-SCHEDULE, AND EVALUATION SETTINGS (MAP@0.5, %). DEFINING ARCHITECTURAL COMPONENTS ARE RETAINED FOR EACH DETECTOR.

Complete detector	mAP@0.5
Oriented R-CNN baseline	50.22
S2A-Net	51.70
ReDet	51.45
ReDet + CSGC	53.55
Oriented R-CNN + CSGC	53.88

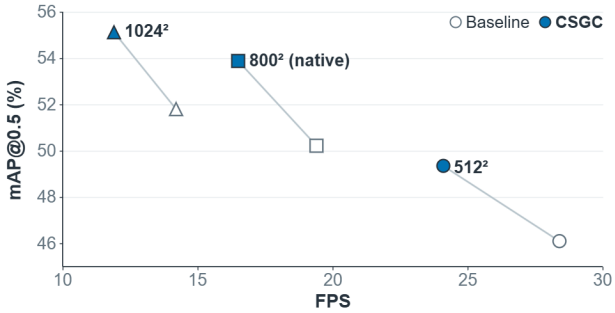


Fig. 7. Paired accuracy–throughput measurements on DOTA-v2.0. Open and filled markers denote the baseline and CSGC; circles, squares, and triangles encode 512^2, native 800^2, and 1024^2 inputs. Each gray segment links methods evaluated at the same resolution and does not represent a cross-resolution trajectory. FPS is measured per crop at batch size 1 with FP16 on one RTX 3090 after 200 warm-up and 500 timed iterations. FPS uses the same per-crop timing convention for the compared configurations.

31.02 M baseline. At 1024^2, computation rises from 208.4 to 215.5 GFLOPs (3.41%), while throughput decreases from 14.2 to 11.9 FPS (16.2%). At the native 800^2 crop resolution, the corresponding throughput change is 19.4 to 16.5 FPS (14.9%).

The component accounting reproduces the full-model increment. CSSC adds 4.7 and 2.9 GFLOPs at 1024^2 and 800^2, respectively; their ratio of 0.617 closely follows the input-area ratio $(800/1024)^2 = 0.610$. Under the fixed proposal count, SEGC adds 2.4 GFLOPs at either resolution because it replaces the two 3×3 RoI-head convolutions. The parameter increments of CSSC and SEGC are 0.84 and 0.07 M, respectively; the latter is net of the 1.18 M convolutional block replaced by the 1.25 M SEGC operator. The 0.004 M FIGAS head is removed at inference and is therefore excluded from the reported inference totals.

Measured throughput does not scale proportionally with the FLOP count. At 800^2, a 3.6% increase in FLOPs accompanies a 14.9% decrease in FPS. FLOP counts and aggregate throughput alone do not distinguish memory traffic, kernel efficiency, launch overhead, or other implementation factors; such attribution requires operator-level profiling. During training, FIGAS increases epoch time by approximately 12.5%, primarily because Eq. (11) evaluates two perturbed polygon IoUs, while peak mixed-precision memory remains 18.2 GB. Runtime and memory figures are empirical properties of the reported implementation rather than asymptotic estimates.

The factorial configurations provide distinct accuracy–throughput operating points. FIGAS alone adds 1.46 mAP without inference overhead, whereas CSSC + FIGAS adds 2.90 points with a 5.7% FPS reduction; the full configuration adds 3.66 points with a 14.9% reduction. Table VIII and Fig. 7 extend the comparison across three input sizes. The CSGC margin remains between 3.25 and 3.66 points, and the relative FPS reduction ranges from 14.9% to 16.2%. Aggregate accuracy rises with resolution for both configurations, but the available results contain no size-stratified evidence identifying which object scales account for the increase. All FPS values describe per-crop throughput; full-scene latency additionally depends on crop generation and merging.

For deployment, full CSGC at $K = 1$ reaches 53.30 mAP@0.5 at 17.5 FPS, compared with 53.88 and 16.5 FPS at $K = 4$, both at 800×800 input (Table VII). Thus $K = 1$ trades 0.58 percentage points for 1.0 FPS, approximately 6.1% higher throughput. These are full-model results, distinct from the CSSC-only rotation-count ablation.

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TABLE V
SELECTED PUBLISHED RESULTS OF TRANSFORMER-BASED ORIENTED DETECTORS. BACKBONES, SCHEDULES, DATA SPLITS, AND EVALUATION CONVENTIONS FOLLOW THE CITED SOURCES ($1\times = 12$ EPOCHS; $3\times = 36$ EPOCHS). VALUES ARE NOT DIRECTLY COMPARABLE WITH TABLE II. "—" DENOTES A RESULT NOT REPORTED IN THE CITED SOURCE; ^{v12} DENOTES THE VOC12 METRIC.

Method	Backbone	Schedule	DOTA-v1.0	DOTA-v2.0	DIOR-R	HRSC2016
ARS-DETR [58]	R-50	$3\times$	74.16	—	66.12	—
OrientedFormer [59]	R-50	$1\times$	75.37	54.27	67.28	90.17
RO^2-DETR [60]	HGNetv2	reported	77.60	52.90	72.26	97.47^{v12}

TABLE VI
LOCALIZATION ACCURACY UNDER VOC07 AND COCO EVALUATION (%). COCO COLUMNS USE EXACT POLYGON IOU WITH MAXDETS = 100. DISPERSION DENOTES THE FIVE-SEED SAMPLE SD WHERE SEED-LEVEL RECORDS ARE AVAILABLE. COCO VALUES FOR SEGC ALONE ARE THREE-SEED MEANS; THE CORRESPONDING VOC07 VALUE RESTATES THE FIVE-SEED FACTORIAL MEAN IN SEC. IV-D1. DATASET SPLITS FOLLOW SEC. IV-A; "—" DENOTES AN UNAVAILABLE MEASUREMENT.

Benchmark	Config.	VOC07	COCO protocol		
		AP_{50}	AP_{50}	AP_{75}	mAP@0.5:0.95
DOTA-v2.0	baseline	50.22 ± 0.14	49.75	28.10	27.85 ± 0.11
	+ SEGC only	51.42	50.90	30.25	29.41
	CSGC	$\mathbf{53.88 \pm 0.10}$	**53.30**	**32.15**	$\mathbf{30.62 \pm 0.14}$
DIOR-R	baseline	64.60	64.15	37.80	36.14 ± 0.12
	CSGC	**66.80**	**66.35**	**40.90**	$\mathbf{38.52 \pm 0.14}$
HRSC2016	baseline	88.30	89.15	74.80	64.80 ± 0.28
	CSGC	**90.70**	**91.04**	**80.10**	$\mathbf{69.15 \pm 0.31}$

TABLE VII
INFERENCE COST ON DOTA-V2.0 AT BATCH SIZE 1, FP16, AND ONE RTX 3090. FLOPS USE FVCORE OVER CONVOLUTION AND LINEAR OPERATORS AT $N_{\text{prop}} = 678$, COUNTING ONE FUSED MULTIPLY–ADD AS ONE FLOP; BOTH INPUT SIZES USE THE SAME EXPORTED GRAPH. FPS IS PER-CROP THROUGHPUT AFTER 200 WARM-UP AND 500 TIMED ITERATIONS. ΔFPS IS RELATIVE TO THE 1024^2 BASELINE. BOLD IDENTIFIES THE FULL CONFIGURATION. THE ROTATION-COUNT BLOCK USES THE FULL MODEL AT 800×800; ACCURACY IS VALIDATION MAP@0.5. "—" DENOTES A MEASUREMENT NOT REPORTED FOR THAT ROW.

Configuration	Params (M)	FLOPs (G)		FPS		ΔFPS	mAP@0.5
		1024^2	800^2	800^2	1024^2		
Module configurations							
Oriented R-CNN (baseline)	31.02	208.4	146.1	19.4	14.2	0.0%	—
+ CSSC	31.86	213.1	149.0	18.3	13.1	−7.7%	—
+ SEGC	31.09	210.8	148.5	17.6	12.4	−12.7%	—
+ FIGAS (training-only)	31.02	208.4	146.1	19.4	14.2	0.0%	—
CSGC (full)	**31.93**	**215.5**	**151.4**	**16.5**	**11.9**	**−16.2%**	—
Full-model rotation-count comparison (800×800)							
CSGC, $K = 1$	—	—	—	17.5	—	—	53.30
CSGC, $K = 4$	—	—	—	16.5	—	—	53.88

TABLE VIII
ACCURACY AND THROUGHPUT ACROSS INPUT RESOLUTIONS ON THE DOTA-V2.0 VALIDATION SPLIT. ACCURACY IS MAP@0.5 (%); FPS IS MEASURED PER CROP AT BATCH SIZE 1 WITH FP16 ON ONE RTX 3090 AFTER 200 WARM-UP AND 500 TIMED ITERATIONS. FPS USES THE SAME PER-CROP TIMING CONVENTION FOR THE COMPARED CONFIGURATIONS. THE 800^2 ROW IS THE NATIVE CROP PROTOCOL, WHEREAS 1024^2 USES TEST-TIME UPSAMPLING.

Input	Baseline		CSGC		ΔFPS
	mAP@0.5	FPS	mAP@0.5	FPS	
512^2	46.10	28.4	49.35	24.1	−15.1%
800^2 (native)	50.22	19.4	53.88	16.5	−14.9%
1024^2	51.80	14.2	55.12	11.9	−16.2%

D. Ablation Studies and Controlled Comparisons

1) Full-Factorial Component Analysis: Table IX reports the complete 2^3 design, allowing each component to be evaluated while the other two stages are held fixed. All eight configurations share the DOTA-v2.0 split, training objective, optimization protocol, and five fixed seeds. Sample SD is reported for the baseline and full configuration. The factorial cells quantify how each component's accuracy difference depends on the other two stages.

Relative to the baseline mean of 50.22, the isolated increments are +1.83 for CSSC, +1.20 for SEGC, and +1.46 for FIGAS. Across the four matched toggles in which each component can be enabled, the conditional differences remain positive: +1.30 to +1.83 for CSSC, +0.76 to +1.20 for SEGC, and +1.00 to +1.46 for FIGAS. No observed toggle reverses direction, although the magnitude depends on which other stages are active.

Among the two-component configurations, CSSC + FIGAS gives the highest mean at 53.12; adding SEGC raises the

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TABLE IX
COMPONENT ANALYSIS (MAP@0.5, %). THE DOTA-V2.0 BLOCK REPORTS THE COMPLETE 2^3 DESIGN AND FIVE-SEED MEANS. THE HRSC2016 TEST BLOCK REPORTS SINGLE-MODULE RESULTS AND THE BASELINE/FULL-MODEL TEST VALUES. CHECKMARKS AND DASHES DENOTE ACTIVE AND INACTIVE COMPONENTS; Δ IS RELATIVE TO THE CORRESPONDING BASELINE.

Configuration	CSSC	SEGC	FIGAS	mAP@0.5	Δ
DOTA-v2.0 validation: factorial analysis					
Baseline	–	–	–	50.22	–
+ CSSC	✓	–	–	52.05	+1.83
+ SEGC	–	✓	–	51.42	+1.20
+ FIGAS	–	–	✓	51.68	+1.46
+ CSSC + SEGC	✓	✓	–	52.88	+2.66
+ CSSC + FIGAS	✓	–	✓	53.12	+2.90
+ SEGC + FIGAS	–	✓	✓	52.58	+2.36
Full CSGC	✓	✓	✓	**53.88**	+3.66
HRSC2016 test: single-module analysis					
Baseline	–	–	–	88.30	–
+ CSSC	✓	–	–	89.10	+0.80
+ SEGC	–	✓	–	89.45	+1.15
+ FIGAS	–	–	✓	89.60	+1.30
Full CSGC	✓	✓	✓	90.70	+2.40

full configuration to 53.88, a further 0.76 points. The full-model increment of 3.66 points is 0.83 below the sum of the three isolated increments (4.49), so the means do not support an additive or synergistic interpretation. The factorial evidence supports a narrower conclusion: the full configuration has the highest reported mean, and every component has a positive conditional difference within the complete design. Section IV-D2 next tests whether the full-model result can be explained by added capacity alone.

The HRSC2016 test block in Table IX extends the single-module comparison beyond DOTA-v2.0. Relative to the 88.30 baseline, CSSC, SEGC, and FIGAS alone reach 89.10, 89.45, and 89.60, with differences of 0.80, 1.15, and 1.30 points. The full model reaches 90.70. These single-module results do not extend the complete factorial interaction analysis to HRSC2016. The FIGAS-only test value 89.60 is distinct from the FIGAS-only validation value 90.70 in Table XIII.

2) Capacity-Matched RoI-Support Perturbation: The factorial configurations do not isolate a capacity-based explanation because their inference architectures differ in size. Table X therefore introduces a capacity-matched perturbation control that retains the complete CSGC architecture, parameter count, FLOPs, training schedule, and seed set while rotating the RoI sampling support by 90°. The comparison tests whether full-model capacity is sufficient to recover the intact result; it is not a pure test of geometric agreement because the intervention also changes the sampled image evidence.

The intervention changes the angle supplied to rotated RoI Align while leaving all box-dependent supervision unchanged. For every proposal $b = (x, y, w, h, \theta)$, sampling uses $\theta + 90°$ with fixed w and h and without le90 re-wrapping; proposal matching, regression targets, evaluation, and the analytical term $\bar{S}_i$ continue to use b. Rotating only the SEGC coordinate basis would yield an invertible reparameterization that a retrained offset branch could absorb, whereas rotating the support places an elongated extraction window across rather than along the proposal axis. The RoI head and the branch producing ψ_i consequently receive perturbed image evidence, while $\bar{S}_i$ remains unchanged. The modified operator is active during training and inference, and the complete model is retrained under the intact protocol.

The capacity-matched model records 51.48 ± 0.09 mAP@0.5, compared with 53.88 ± 0.10 for intact CSGC. The 2.40-point decrease removes 65.6% of the full configuration's 3.66-point improvement over the baseline, despite identical parameter and FLOP counts. The perturbed mean also lies 1.40 points below the feature-conditioned, assignment-blind configuration. The controlled result disfavors a capacity-only account within the reported pipeline. Sampling-support rotation nevertheless changes both geometric agreement and accessible image content, leaving their individual contributions unresolved.

The same support rotation reduces the baseline from 50.22 to 48.90, a 1.32-point decrease, compared with the 2.40-point decrease for full CSGC. Their decrease difference is 1.08 points, a descriptive contrast rather than an isolated geometric effect. With the full detection pathway retained but the gate reading an unconditioned feature copy, accuracy is 50.60, or 3.28 points below full CSGC. Together these controls address the feature-to-assignment linkage while preserving the limitation that support rotation also changes sampled image content.

Figure 8 examines whether the category-level loss under support rotation covaries with an idealized measure of uncovered support. For concentric equal-size rectangles, the intersection after a 90° rotation occupies h^2 of the original wh area; with $\rho = w/h \geq 1$, the retained fraction is $1/\rho$ and the uncovered fraction is $1-1/\rho$. Regressing the intact-minus-perturbed category contrast on the uncovered fraction gives a slope of 4.15 points per unit and $R^2 = 0.89$, with Spearman $r_s = 0.97$; direct regression on ρ gives $R^2 = 0.72$. The full-minus-baseline contrast has a weaker linear association with the same proxy ($R^2 = 0.38$). The proxy and fitted lines summarize observed category means rather than measure removed feature evidence directly.

Category-level departures from the fitted relation show that the geometric proxy is not exhaustive. `Bridge` and `harbor` have the largest positive residuals, whereas `helicopter` shows a smaller loss at a comparable aspect ratio; any explanation based on the spatial distribution of discriminative evidence remains post hoc. The perturbed configuration falls below the baseline in four categories, with the largest decrease for `airport` (42.10 → 41.10). The estimated association between elongation and the fraction of category gain retained after perturbation is $r_s = -0.23$ ($p = 0.35$). Proposal localization, padding, interpolation, background content, category frequency, and appearance remain potential confounders; Sec. V discusses the remaining limitation in separating geometric agreement from sampled image content.

3) Stage-Matched Alternative Modules: Table XI asks whether the proposed stage-specific operators retain an advantage over selected alternatives inserted at the same detector interface. CBAM-L, DCNv2, ARC, and the EARL-based assignment rule are local transplants rather than official end-

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TABLE X
ROI-SUPPORT AND FEATURE-COPY CONTROLS ON DOTA-V2.0 (MAP@0.5, %). THE BASELINE, FULL CSGC, AND FULL-MODEL SUPPORT-ROTATION RESULTS REPORT FIVE-SEED MEANS AND SAMPLE SD. FEATURE CONDITIONING DENOTES CSSC +SEGC AND ASSIGNMENT CONDITIONING DENOTES FIGAS. BOTH SUPPORT CONTROLS SHIFT ONLY THE SAMPLING ANGLE BY 90°. THE FEATURE-COPY CONTROL CHANGES ONLY THE GATE INPUT TO AN UNCONDITIONED ROI-FEATURE COPY AND RETAINS THE FULL DETECTION PATHWAY. *CONDITIONING REMAINS ACTIVE BUT IS INCONSISTENT WITH DOWNSTREAM GEOMETRY.

Configuration	Feature cond.	Assignment cond.	mAP@0.5	Gap to full
Baseline	No	No	50.22 ± 0.14	3.66
Baseline, rotated RoI support	No	No	48.90	4.98
Assignment-aware, feature-blind	No	✓	51.68	2.20
Feature-aware, assignment-blind	✓	No	52.88	1.00
Full CSGC, feature-copy gate	✓	✓	50.60	3.28
Full capacity, rotated RoI support	✓*	✓	51.48 ± 0.09	2.40
Coordinated stages (CSGC)	✓	✓	$\mathbf{53.88 \pm 0.10}$	0

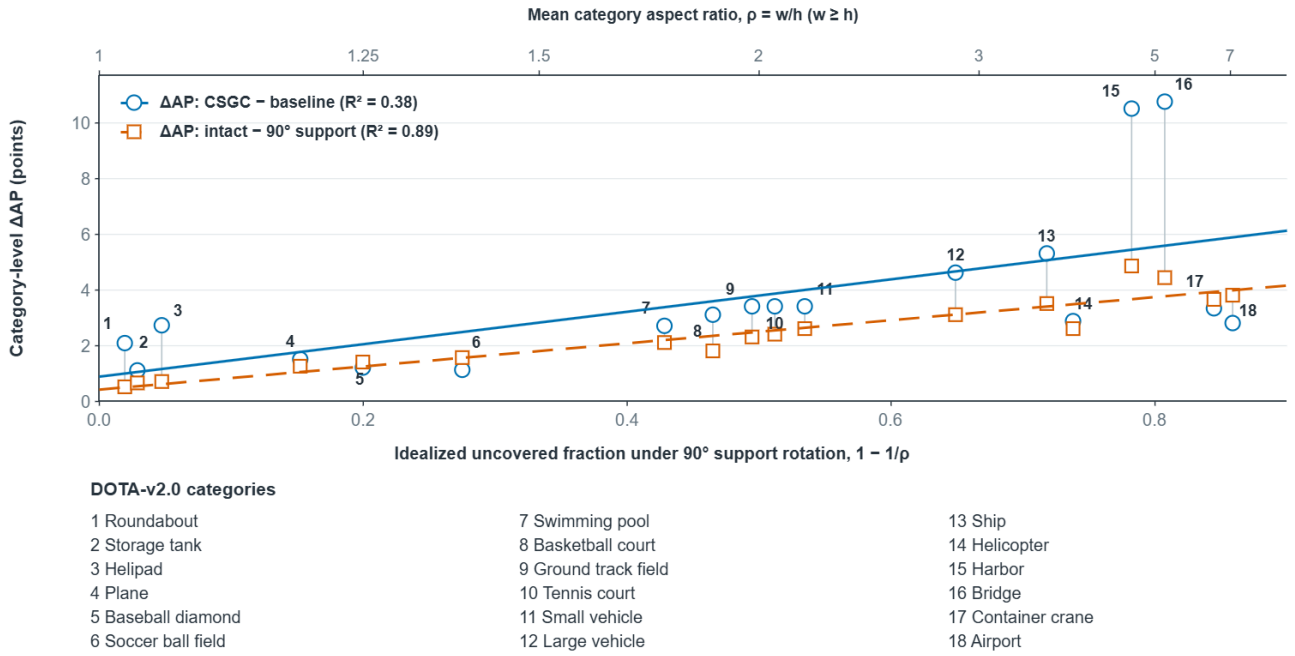


Fig. 8. Category-level AP contrasts and target elongation on DOTA-v2.0. Blue circles show $\mathrm{AP}_{\mathrm{CSGC}} - \mathrm{AP}_{\mathrm{baseline}}$; orange squares show $\mathrm{AP}_{\mathrm{intact}} - \mathrm{AP}_{90^\circ\text{-support}}$ across 18 categories. The lower axis is the idealized uncovered fraction $1 - 1/\rho$, and the upper axis gives the corresponding mean category aspect ratio $\rho = w/h$ ($w \geq h$). Pale vertical segments pair the two contrasts for a category and are not uncertainty intervals. Lines are descriptive ordinary-least-squares fits; neither the proxy nor the associations establish causality.

to-end reproductions of the cited detectors. The repetition protocols are not uniform across all rows, so cross-row differences are treated as descriptive means rather than seed-matched estimates.

At the FPN interface, CBAM-L enlarges the channel width of per-level CBAM [48] to approximate the parameter increment of CSSC (+0.82 versus +0.84 M). The corresponding means are 51.15 and 52.05, a 0.90-point difference. Parameter matching narrows a capacity explanation but does not equalize receptive field, operator structure, or optimization behavior. At the assignment interface, the EARL transplant records 51.88, whereas the three-seed FIGAS remeasurement records 52.11; the 0.23-point difference crosses repetition protocols and is therefore not used to rank the two rules.

At the RoI interface, DCNv2 and SEGC both predict $2N$ offset scalars per position from the same input, using Cartesian and radial–tangential coordinates, respectively. Their reported means differ by only 0.07 points (51.35 versus 51.42), again across three- and five-seed summaries. ARC records the highest mean among the selected RoI alternatives at 52.02, so Table XI does not support a claim that SEGC is the most accurate RoI operator. The separately summarized three- and five-seed FIGAS means differ by 0.43 points (52.11 versus 51.68), illustrating why smaller cross-protocol gaps cannot resolve an ordering. Feature-space rotation-consistency measurements address a different question and are reported separately in Sec. IV-E.

The simultaneous CBAM–DCNv2–EARL replacement reaches 52.10 under the shared validation setting, compared with 50.22 for the baseline and 53.88 for CSGC. The 1.78-point difference favors the complete conditioning configuration in this comparison. It does not imply that SEGC is individually

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superior to ARC or DCNv2, nor does it isolate a causal contribution of the coupling constraint.

4) Design Choices and Sensitivity Analyses: **Number of rotated feature copies.** Table XII isolates the number of rotated copies in the CSSC-only configuration. The reported mean rises from 51.28 at $K = 1$ to 52.05 at $K = 4$ and 52.11 at $K = 8$, corresponding to increments of 1.06, 1.83, and 1.89 points over the baseline. The $K = 1$ setting retains the attention pathway without rotation aggregation, whereas $K = 4$ and $K = 8$ sample the half-turn interval at progressively finer angular spacing.

The $K = 1$ configuration, which retains the unrotated attention-and-residual pathway, improves mAP@0.5 by 1.06 points over the baseline. Rotation aggregation at $K = 4$ provides a further 0.77-point improvement in the reported mean. Coordinate Attention reaches 51.60, compared with 51.15 for CBAM-L and 52.05 for CSSC at $K = 4$. The corresponding 0.45- and 0.90-point differences are descriptive and do not partition the gain into independent causal contributions.

The $K = 1$ and $K = 8$ entries use the same three-seed protocol and differ by 0.83 points, providing the direct comparison between unrotated attention and rotation aggregation. The $K = 4$ and $K = 8$ entries use different repetition protocols; their 0.06-point mean difference alone establishes neither equality nor saturation. Rotated-copy cost scales linearly with K, making $K = 4$ the lower-cost default among the two aggregated settings without identifying a statistically resolved optimum.

FIGAS parameters and gate controls. The default parameters were selected on the DOTA-v2.0 validation set from $\alpha \in \{0, 0.15, 0.30, 0.45, 0.60\}$ and $\beta \in \{0.3, 0.5, 0.7\}$. The same grid was evaluated on the HRSC2016 validation split to examine transfer to a benchmark dominated by elongated targets, while the HRSC2016 test result in Table II remained outside hyperparameter selection. The selected pair, $(\alpha, \beta) = (0.3, 0.5)$, lies in a locally stable region on both validation sets: the eight neighboring HRSC2016 cells span 90.54–90.70 mAP@0.5, and the DOTA-v2.0 grid exhibits similarly small local variation. Values of $\alpha \geq 0.6$ produce oscillating regression losses, especially for extreme-aspect-ratio categories such as harbors and bridges. The instability is compatible with overly permissive positive selection, although this hyperparameter comparison does not isolate its cause. Across the two validation settings, the selected pair remains competitive without dataset-specific retuning.

Under the default (α, β) setting, the median number of positive candidates per DOTA-v2.0 development image increases from 18.5 under the baseline assignment rule to 21.2 under FIGAS. The count records the operational consequence of threshold relaxation and does not establish that every additional positive is correctly assigned.

Table XIII reports the perturbation-angle and attenuation-gate controls. Changing δ from 5° to 20° alters the FIGAS-only mean by at most 0.19 points over the fourfold range. Constant gates yield 51.05 for $\psi_i \equiv 1$ and 51.24 for $\psi_i \equiv 0.52$, while shuffling the learned scores across candidates yields 50.88; the learned gate reaches 51.68. The ordering is consistent with candidate-specific gate values carrying information beyond their marginal magnitude. No control establishes that ψ_i measures feature quality or that upstream conditioning causes the observed difference.

The attenuation-gate rows use the FIGAS-only configuration, with CSSC and SEGC inactive; the gate is therefore computed from an unconditioned feature stream. The comparison quantifies a learned attenuation gate within FIGAS rather than the upstream feature-to-assignment linkage described in Sec. II-D.

The full-model training diagnostic in Fig. 11 complements these FIGAS-only controls. Over the 12-epoch schedule, the mean attenuation gate decreases from approximately 0.54 to 0.32 and does not approach zero within the 12-epoch schedule. This observation describes finite-schedule behavior and does not make the gate a calibrated feature-quality measure. The raw sensitivity distribution contains $1{,}000$ candidate–ground-truth pairs; with $\rho_{\max} = 12$, $S_{\max} = \frac{1}{2}(12 + 12^{-1}) = 6.0417$ rad^{-1}. The observed maximum $S_i \approx 2.63$ rad^{-1} corresponds to a maximum normalized sensitivity of approximately 0.4353, so no pair in this diagnostic sample reaches the cap.

E. Diagnostic and Qualitative Analysis

Figure 9 separates the rotation-consistency measurements into grid symmetries, diagonal rotations, and their equal-weight eight-angle mean. The resulting diagnostic characterizes feature behavior independently of the accuracy tables.

Equivariance error. Under the shared diagnostic settings in Sec. III-C, the eight-angle mean errors are 0.0405 for SEGC and 0.0390 for the full configuration. Figure 9 shows lower feature-space error for SEGC than for ARC, whereas Table XI gives higher mAP@0.5 for ARC (52.02 versus 51.42). The diagnostic and AP measure different outcomes; their orderings need not coincide. Separate training and simultaneous changes in representation prevent these observations from establishing that reduced consistency error causes improved detection accuracy.

The strict identity reference $f(\mathbf{x}) = \mathbf{x}$ has zero error under Eq. (10). Operator comparisons use the errors defined by that equation without reference subtraction.

Gain versus aspect ratio. The small-angle self-IoU analysis of Sec. III-D motivates the hypothesis that observed gains may vary with elongation. A descriptive DOTA-v2.0 analysis uses the per-category difference between the full configuration and the matched baseline across all 18 categories; the unweighted mean reproduces the aggregate $+3.66$-point difference by construction. Stratum- and category-level summaries address complementary aspects of the association.

Aspect-ratio strata exhibit a monotone sequence of mean differences: $+1.62$ for $\rho \leq 1.5$ (six categories), $+3.20$ for $1.5 < \rho \leq 2.2$ (five), $+4.26$ for $2.2 < \rho \leq 4.0$ (three), and $+6.85$ for $\rho > 4.0$ (four). Fitting the four stratum means against the within-stratum average of the idealized proxy $1 - 1/\rho$ gives $\Delta\text{AP} = 6.63(1 - 1/\rho) + 0.45$ with $R^2 = 0.87$. For real categories, the proxy is not a measured fraction of missed target or feature support. The fit therefore describes four aggregated strata ($n = 4$), which can exhibit a stronger apparent association than the underlying category-level data.

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TABLE XI
STAGE-LOCAL AND JOINT MODULE COMPARISONS ON DOTA-V2.0 VALIDATION (MAP@0.5, %). CBAM-L, DCNV2, AND FIGAS REMEASUREMENT ROWS USE THREE SEEDS (3407, 42, 2023); THE OTHER STAGE-LOCAL ROWS AND THE JOINT-BLOCK BASELINE/FULL-MODEL ROWS REPORT FIVE-SEED MEANS. THE JOINT REPLACEMENT CHANGES ALL THREE INTERFACES TOGETHER. DIFFERENCES BETWEEN REPORTED RESULTS ARE DESCRIPTIVE; Δ IS RELATIVE TO THE SHARED BASELINE.

Stage	Method	Runs	mAP@0.5	Δ
FPN modulation	CBAM-L	3	51.15	+0.93
	CSSC	5	52.05	+1.83
RoI convolution	Standard conv.	5	50.22	0.00
	DCNv2	3	51.35	+1.13
	ARC	5	52.02	+1.80
	SEGC	5	51.42	+1.20
Positive-sample assignment	EARL	5	51.88	+1.66
	FIGAS	3	52.11	+1.89
	Joint three-stage comparison			
Reference	Shared baseline	5	50.22	0.00
Joint replacement	CBAM + DCNv2 + EARL	—	52.10	+1.88
Joint conditioning	Full CSGC	5	53.88	+3.66

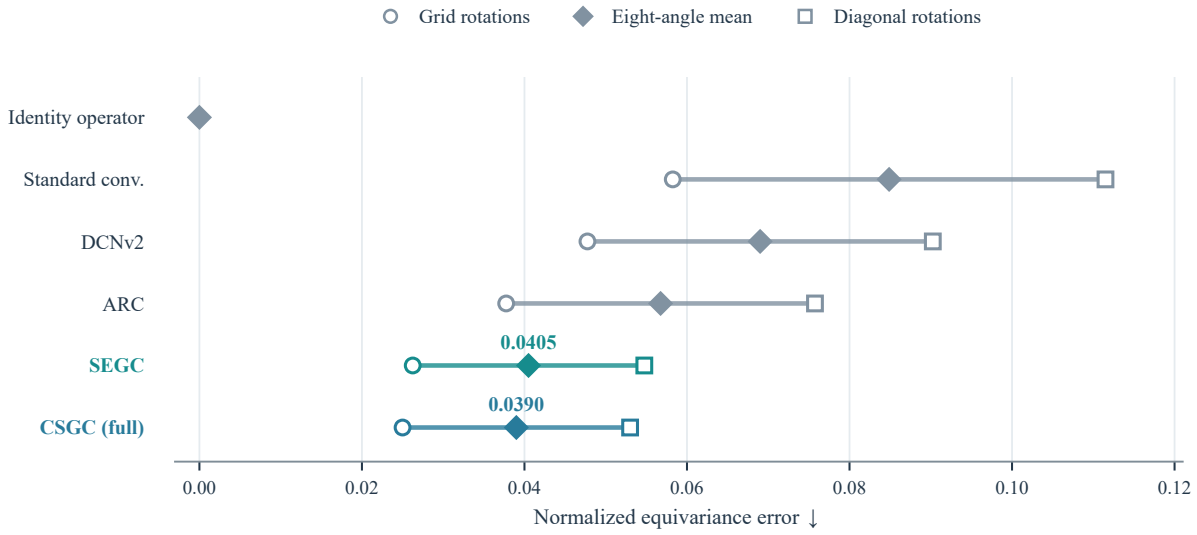


Fig. 9. RoI-level rotation-consistency error. Circles and squares denote means over four grid and four diagonal rotations; diamonds denote the equal-weight eight-angle mean over 1,000 matched proposals. Identity denotes $f(\mathbf{x}) = \mathbf{x}$, with zero error under Eq. (10). Lines connect group means and are not error bars. Lower error describes feature-space consistency; detection accuracy is evaluated separately.

At the category level, mean aspect ratio and gain have a Spearman correlation of $r_s = 0.72$ ($n = 18$; nominal $p < 0.001$), whereas the ordinary least-squares fit shown by blue circles in Fig. 8 reaches only $R^2 = 0.38$. Category summaries from one model pair make the nominal p-value descriptive rather than confirmatory. Frequency, object size, and appearance remain potential confounders. Harbor and bridge have the largest AP gains (+10.50 and +10.75), while airport and container-crane show smaller gains despite high aspect ratios (+2.80 and +3.33). This category-level pattern motivates a geometric interpretation.

Accuracy across angle strata. Table XIV stratifies DOTA-v2.0 accuracy by unsigned angle magnitude $|\theta| \in [0°, 90°)$, discarding the sign of the native le90 angle. Baseline accuracy decreases monotonically toward the boundary-adjacent stratum, with a 4.80-point gap between the first and last bins. Under CSGC, the corresponding gap narrows to 3.10 points and the cross-stratum standard deviation decreases from 1.97 to 1.27. The largest difference from the baseline, +4.80 points, occurs in the weakest baseline stratum. Because the table evaluates the full configuration, the result locates an aggregate difference without attributing it to a single component. The angle term in Eq. (15) is shared by both configurations (Sec. III-E); identifying the source of the boundary-adjacent difference would require a stratified component ablation.

Boundary conditions and long-tail behavior. Category-level results delimit settings with smaller reported mean gains. For near-isotropic targets, $\frac{1}{2}(\rho + \rho^{-1})$ approaches its minimum as $\rho \to 1$, causing the geometric term of FIGAS to approach uniformity; the lowest aspect-ratio stratum records a +1.62-point mean difference. Near the angular boundary, the re-wrapping in Eq. (11) keeps the perturbed rectangle well

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TABLE XII
ROTATION-COUNT AND REFERENCE-ATTENTION ANALYSIS ON DOTA-V2.0 VALIDATION (MAP@0.5, %). CSSC USES K EQUALLY SPACED ANGLES IN $[0°, 180°)$: $K = 1$ USES $\{0°\}$; $K = 4$ AND $K = 8$ USE $45°$ AND $22.5°$ INCREMENTS. $K = 1$ AND $K = 8$ REPORT THREE-SEED MEANS; $K = 4$ REPORTS ITS FIVE-SEED FACTORIAL MEAN. CBAM-L USES THREE SEEDS. Δ IS RELATIVE TO THE 50.22 BASELINE; RELATIVE BRANCH COST APPLIES TO CSSC.

Configuration	K	mAP@0.5	Δ	Rel. cost
CSSC rotation-count configurations				
CSSC	1	51.28	+1.06	1×
CSSC (default)	4	52.05	+1.83	4×
CSSC	8	52.11	+1.89	8×
Reference attention configurations				
Baseline	—	50.22	0.00	—
CBAM-L	—	51.15	+0.93	—
Coordinate Attention	—	51.60	+1.38	—

TABLE XIII
SENSITIVITY OF FIGAS (FIGAS-ONLY CONFIGURATION; MAP@0.5, %; DOTA-V2.0 BASELINE 50.22). SWEPT AND PROBED SETTINGS ARE THREE-SEED MEANS; THE DEFAULT ROW RESTATES THE FIVE-SEED FACTORIAL MEAN OF TABLE IX. "SHUFFLED" PERMUTES THE LEARNED SCORES ACROSS CANDIDATES, PRESERVING THEIR DISTRIBUTION WHILE DESTROYING THEIR CORRESPONDENCE. Δ IS RELATIVE TO THE DOTA-V2.0 BASELINE. THE DEFAULT-ROW HRSC (VAL) VALUE AND THE FULL-MODEL HRSC TEST FIGURE OF TABLE II (90.70) ARE DISTINCT MEASUREMENTS THAT COINCIDE NUMERICALLY.

Factor	Setting	DOTA (val)	HRSC (val)	Δ
δ	5°	51.54	90.58	+1.32
	10° (default)	**51.68**	**90.70**	+1.46
	20°	51.49	90.62	+1.27
ψ_i	$\equiv 1$ (saturated)	51.05	—	+0.83
	$\equiv 0.52$ (fixed gate)	51.24	—	+1.02
	shuffled	50.88	—	+0.66
	learned (full FIGAS)	**51.68**	—	+1.46

TABLE XIV
MAP@0.5 (%) ON DOTA-V2.0 STRATIFIED BY UNSIGNED ANGLE MAGNITUDE. σ IS THE POPULATION SD ACROSS THE THREE STRATA, A DESCRIPTIVE UNIFORMITY INDEX. EACH STRATUM IS SCORED ON ITS OWN SUBSET OF THE GROUND TRUTH, SO THE OVERALL MAP IS NOT A CONVEX COMBINATION OF THE THREE.

Method	$[0°, 30°)$	$[30°, 60°)$	$[60°, 90°)$	σ
Oriented R-CNN (baseline)	52.10	49.20	47.30	1.97
CSGC	**55.20**	**53.70**	**52.10**	**1.27**
Δ	+3.10	+4.50	+4.80	−0.70

defined, while discrete polygon-IoU evaluation may be least smooth; the boundary-adjacent stratum remains weakest in absolute mean accuracy. Both observations are descriptive rather than component-level attributions.

Table XV reports the five DOTA-v2.0 categories with the fewest training instances. Their mean rises from 39.42 to 42.18 (+2.76), below the overall +3.66-point difference. Container-crane, the rarest category, improves by +3.33 points from 132 training instances. Category means cannot identify contributions from the geometric term, attenuation gate, or class frequency; rare-class behavior therefore remains unresolved at the component level.

Qualitative detection results. Figure 10 complements the aggregate metrics with scene-level outputs in dense, elongated, and large-object settings.

TABLE XV
THE FIVE DOTA-V2.0 CATEGORIES WITH THE FEWEST TRAINING INSTANCES (AP@0.5, %; FIVE-SEED MEANS). "INST." IS THE NUMBER OF TRAINING INSTANCES.

Category	Inst.	Baseline	CSGC	Δ
container-crane	132	14.82	**18.15**	+3.33
helipad	176	28.50	**31.22**	+2.72
airport	243	42.10	**44.90**	+2.80
helicopter	730	51.33	**54.20**	+2.87
roundabout	748	60.35	**62.43**	+2.08
Mean	—	39.42	**42.18**	+2.76

Figure 10 uses four fixed validation scenes selected from ground truth, identical 800×800 crops, and the same 160×160 inset region with $2\times$ magnification in each row. All detector columns use LSKNet-S, seed 3407, and the final 12-epoch checkpoint; the qualitative comparison shares the training split, batch size, optimizer, learning-rate schedule, and initialization, while retaining method-specific heads. The roundabout, harbor, athletics-track, and bridge rows show localized detection differences and failure cases under a common display convention. These selected scenes provide qualitative evidence; quantitative conclusions rely on the corresponding performance tables.

V. DISCUSSION AND LIMITATIONS

Empirical scope. Evidence spans four benchmarks for the complete configuration, supplemented by complete-detector comparisons and transfer to ReDet. Single-module effects are additionally reported on HRSC2016, but the complete factorial design, support-rotation controls, and detailed stage-matched alternatives remain concentrated on DOTA-v2.0. Component interactions across other benchmarks remain unresolved. DOTA validation informed configuration selection; the separate official-test results use frozen hyperparameters. The selected defaults are transferred to DIOR-R and HRSC2016, with HRSC2016 test reserved from selection. The matched assignment-stage alternative remains EARL; prediction-conditioned strategies reviewed in Sec. II-C are not evaluated in this pipeline.

Comparison conventions. Matched-pipeline and DOTA-v2.0 factorial comparisons use five-seed means. The CBAM-L, DCNv2, and FIGAS remeasurements use three-seed means. Tables distinguish the evaluated configurations and summary conventions; between-configuration differences describe the corresponding reported results.

Mechanistic interpretation. The baseline support-rotation and unconditioned feature-copy controls examine sampling-support effects and the feature-to-assignment linkage. Support rotation nevertheless changes both geometric agreement and sampled image evidence, so its separate effects remain unresolved. The gate trajectory shows no approach to zero during the 12-epoch schedule, while neither it nor the feature-copy contrast establishes a calibrated feature-quality interpretation. Reductions in rotation-consistency error and improvements in

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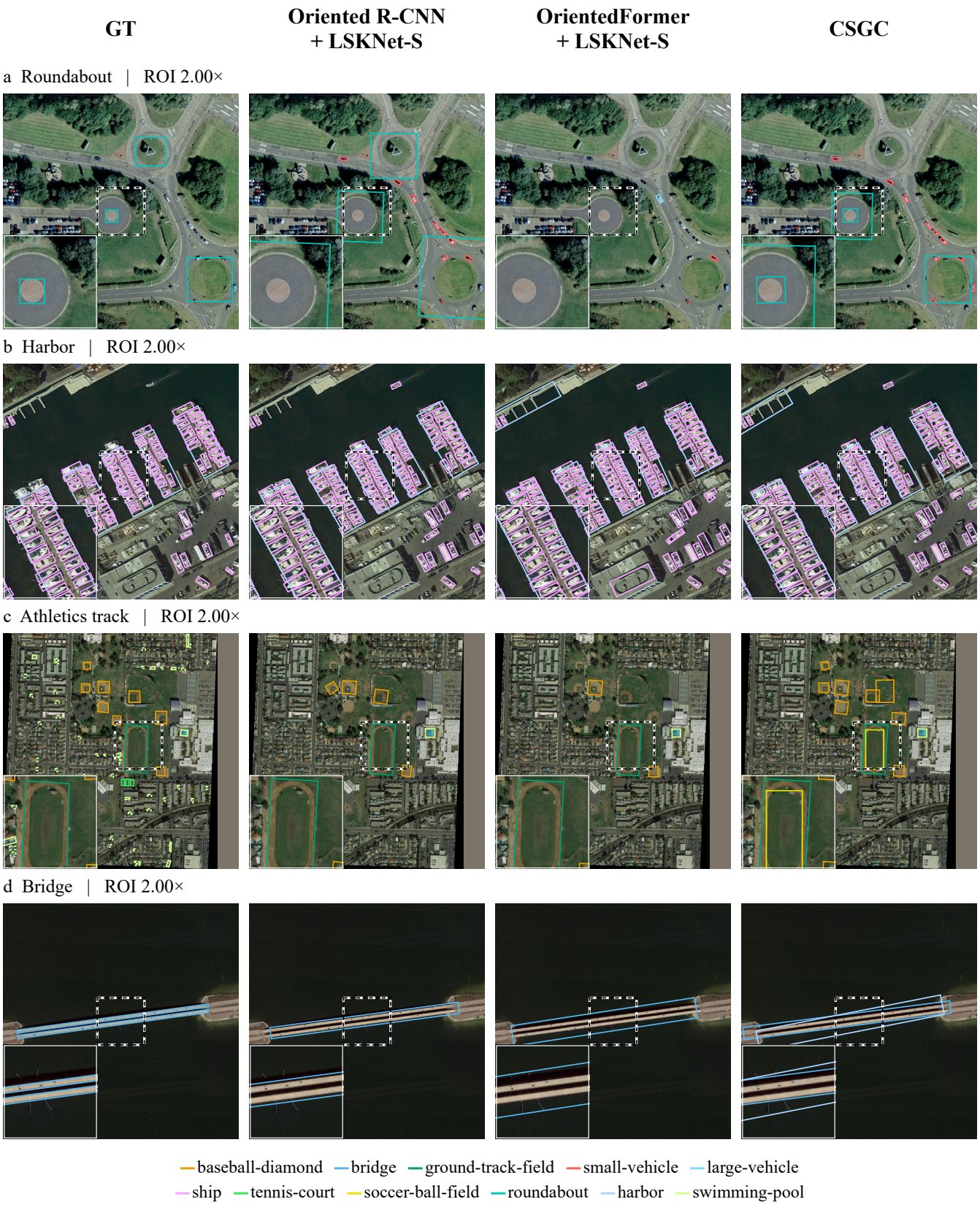


Fig. 10. Qualitative comparison on DOTA-v1.0 validation. Columns: ground truth, Oriented R-CNN+LSKNet-S, OrientedFormer+LSKNet-S, and CSGC. Rows: roundabout, harbor, athletics track, and bridge. Crops, inset regions, class colors, line widths, and the display threshold (score ≥ 0.30) are shared.

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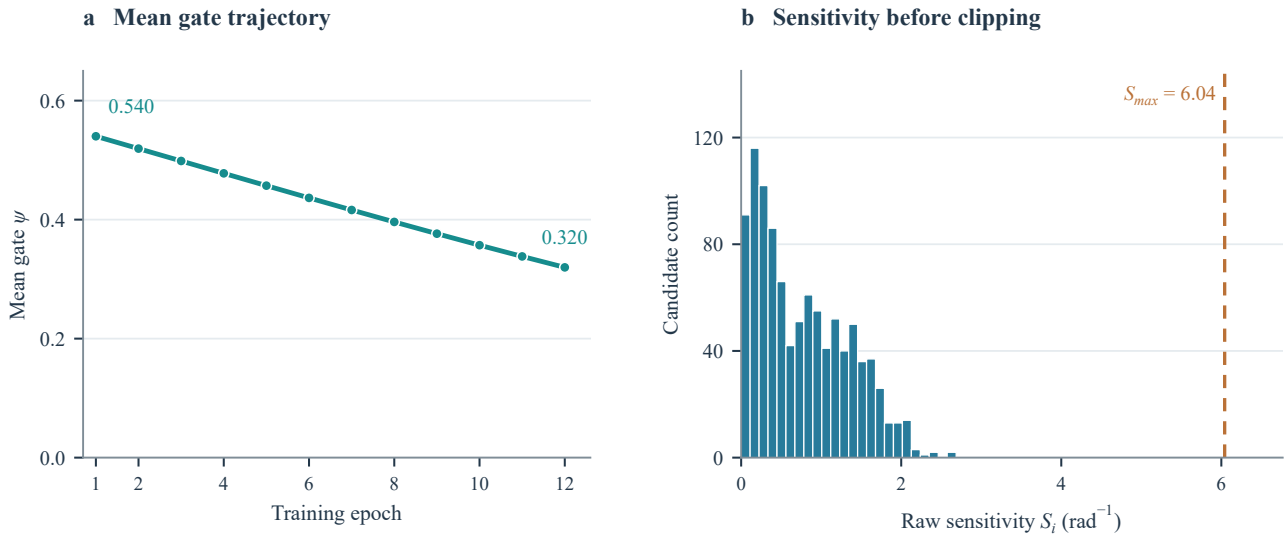


Fig. 11. Mean attenuation-gate trajectory and raw angular-sensitivity distribution. (a) Mean gate values over the 12-epoch training schedule. (b) Raw sensitivities of 1,000 candidate–ground-truth pairs before clipping, with $\delta = \pi/18$. The dashed line marks $S_{\max} = 6.04$ rad^{-1} at $\rho_{\max} = 12$. The observed maximum is approximately 2.63 rad^{-1}; no pair in this diagnostic sample reaches the cap.

AP are co-occurring measurements and do not establish a causal relationship. In particular, SEGC has lower diagnostic error than ARC but lower mAP@0.5; the evidence does not show that reducing the diagnostic error causes an accuracy increase.

Qualitative evidence. Figure 10 uses fixed scenes, aligned crops and insets, and a shared display convention under the comparison settings in Sec. IV-E. Selected scenes provide qualitative examples and do not estimate dataset-level error rates or establish an aggregate ranking against OrientedFormer.

VI. Conclusion

CSGC coordinates geometric conditioning across feature-pyramid modulation, proposal-aligned RoI sampling, and training assignment. On the official DOTA-v1.0 and DOTA-v2.0 Task 1 servers, AP_{50} reaches 77.58% and 53.18%, exceeding the shared baselines by 2.42 and 2.80 points. The DIOR-R and HRSC2016 test improvements are 2.20 and 2.40 points; the HRSC2016 mAP@0.5:0.95 difference is 4.35 points. The validation ablations show positive conditional component differences under the shared protocol. The joint replacement, feature-copy and support controls, HRSC single-module analysis, and ReDet transfer extend the evidence within the evaluated architectures and settings.

The 90° RoI-support perturbation retains the full architecture but reduces its validation accuracy by 2.40 points, compared with a 1.32-point decrease for the baseline. The contrast and feature-copy control constrain capacity-only and feature-source explanations while preserving the sampled-content limitation. The results do not establish detector-independent behavior or a causal link between feature-space consistency and AP.

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